Electronic Supplementary Material for "Do Visual-Selection Operations Track Technical Quality? A Controlled Audit for Multi-Video Summarization"

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This supplement provides operation provenance, implementation detail, complete results and sensitivity analyses supporting the main article, which cites these tables as Table S1, Table S2, and so on.

Abbreviations used in the tables: SRCC, Spearman rank correlation coefficient; PLCC, Pearson linear correlation coefficient; RMSE, root mean square error; OR, outlier ratio; NR-IQA, no-reference image quality assessment; DOVER-T, the technical branch of DOVER.

Table S1 In-domain collection: topics of Brazilian television news, with the number of source videos, segment slots, redundancy clusters and segment duration range. *Slots* counts cluster memberships; because two segments belong to two clusters each, the 33 slots correspond to 31 distinct segments.

Topic	Videos	Slots	Clusters	Duration (s)	Status
bebe_real	3	5	2	8–41	Included
brumadinho	3	6	3	13–188	Included
fone_canc._ruido	3	6	3	16–30	Included
isencao_imposto	3	6	3	25–30	Included
mounjaro	3	4	1	8–25	Included
ucrania	3	6	3	25–29	Included
esquema_pcc	3	–	–	–	Excluded[†]
Total	**18**	**33**	**15**	8–188	—

[†]98 SIFT descriptors $< K = 300$ K-means centroids, so the HSMVideoSumm visual-word score cannot be built for this topic. **The exclusion is a property of that scoring path, not of the topic**: this block is the auxiliary 15-cluster set, whose vocabulary is fitted over the whole topic at ten uniformly spaced frames per segment, which is the configuration that fails here. The 42-cluster operational set below includes the topic because the HSMVideoSumm score runs there under a different configuration: the vocabulary is fitted over the segments of each cluster separately, at one frame per second. The selections that path produced for these three clusters, with their scores, are in `jusante_sift.json` of the reproducibility package. Slots exceed distinct segments in `brumadinho` and `isencao_imposto`, where one segment belongs to two clusters.

Table S1 (continued). Operational 42-cluster set used in the primary application analysis.

Topic	Clusters	Slots	Distinct segments
acordo_mariana	1	2	2
aumento_cafe	1	4	4
bebe_real	2	5	5
bolsonaro_trump	3	6	6
brasil_omc	1	2	2
brumadinho	3	6	5
catedral_notre_dame	2	5	4
cessar_fogo_ucrania	3	6	4
esquema_pcc	3	6	6
fone_cancelamento_ruido	3	6	6
fraude_inss	3	6	4
google_huawei	3	6	4
isencao_imposto	3	6	5
julgamento_bolsonaro	3	6	5
mounjaro	1	4	4
recorde_calor_2024	2	5	5
rio_celular	1	2	2
ucrania	3	6	6
ultraprocessados	1	4	4
Total	**42**	**93**	**83**

Table S2 Illustration of the sensitivity axis on a single broadcast frame and four degradations, after applying each method's declared polarity. The pattern this small case shows is the one Table S3 then measures over 25 distortion types and 10,125 images; nothing is concluded from this table alone. Cells are shaded by outcome: expected order, partial, violated.

Method	Pol.	Original	Blur	Noise	Dark	JPEG	Outcome
QualiCLIP	↑	0.341	0.155	0.167	0.196	0.163	as expected
LIQE	↑	1.294	1.056	1.000	1.023	1.023	as expected
BRISQUE	↓	42.8	90.0	66.4	59.1	78.2	as expected
NIQE	↓	4.53	11.99	15.79	6.18	9.47	as expected
Keypoint density	↑	2598	104	**4543**	4	2799	partial
DINOv2 ViT-S/14	↑	14.69	14.86	14.86	14.90	15.14	violated
ResNet-50	↑	15.62	**18.11**	15.00	15.97	15.28	violated

Values in bold are the ones that violate the expected order.

Table S2 (continued). Exact parameters of the constructed broadcast degradations.

Type	Levels 1–5
Gaussian blur	Kernel width $\{3, 5, 9, 15, 23\}$ pixels
White noise	Gaussian $\sigma = \{5, 12, 22, 35, 50\}$; deterministic seeds 1001–1005
Darkening	RGB multiplier $\{.80, .62, .46, .32, .20\}$ with clipping
JPEG	OpenCV quality $\{40, 25, 15, 9, 5\}$
H.264	libx264 CRF $\{30, 36, 41, 45, 50\}$, preset `veryfast`
Rescale	Downscale factors $\{.75, .55, .40, .28, .18\}$, then restore original size
Motion blur	Horizontal kernel width $\{5, 9, 15, 23, 33\}$ pixels
Desaturation	HSV saturation multiplier $\{.75, .55, .35, .18, .05\}$

Each segment contributes at most 30 uniformly spaced frames. The seeded 20-segment simple random sample represents 13 topics: one segment each from `aumento_cafe`, `bolsonaro_trump`, `cessar_fogo_ucrania`, `esquema_pcc`, `isencao_imposto`, `mounjaro` and `rio_celular`; two each from `bebe_real`, `google_huawei`, `julgamento_bolsonaro`, `recorde_calor_2024` and `ucrania`; and three from `fone_cancelamento_ruido`.

Table S3 Sensitivity to distortion level on KADID-10k. Direction-aligned SRCC is computed within each of the 81 reference images across five levels and then averaged by distortion type and across the 25 types. Intervals block-bootstrap the 81 references (10^4 resamples, seed 42). The last column counts types whose reference-averaged SRCC exceeds 0.50.

Method	Mean ρ^*	95% block CI	Types > 0.5
Q-Align	**0.880**	[0.874, 0.886]	**23/25**
QualiCLIP	**0.849**	[0.840, 0.857]	**22/25**
CLIP-IQA+	**0.808**	[0.795, 0.821]	**22/25**
LIQE	**0.807**	[0.797, 0.817]	**22/25**
MUSIQ	**0.733**	[0.720, 0.747]	**20/25**
TOPIQ	**0.721**	[0.702, 0.737]	**20/25**
HyperIQA	0.673	[0.653, 0.692]	**19/25**
DBCNN	0.668	[0.642, 0.692]	**19/25**
ARNIQA†	0.668	[0.652, 0.683]	**20/25**
BRISQUE	0.657	[0.628, 0.684]	**19/25**
CLIP-IQA	0.645	[0.612, 0.676]	**20/25**
TReS†	0.620	[0.598, 0.640]	**18/25**
NIQE	0.607	[0.587, 0.625]	**19/25**
SI	0.251	[0.219, 0.282]	13/25
Keypoint density	0.079	[0.033, 0.128]	7/25
SIFT+BoVW	−0.077	[−0.124, −0.030]	5/25
DINOv2	−0.064	[−0.136, 0.009]	1/25
ResNet-50	−0.297	[−0.364, −0.225]	0/25

†ARNIQA and TReS are reported on this axis as coverage extensions but are not treated as complete-case methods in the screening rule because the required CLIVE evaluations were not available.

Table S4 P.1401 point report after the logistic mapping. PLCC measures linearity, RMSE accuracy and the outlier ratio (OR; lower is better) consistency; RMSE* is the companion error after discounting the 95% confidence interval of the subjective score. OR saturates on KonIQ-10k because its per-image subjective intervals are narrow, but remains discriminative on KADID-10k. Analytic 95% intervals for PLCC, RMSE, RMSE* and OR are included in the released machine-readable results. A dagger marks evaluation on training data.

Method	KonIQ-10k				KADID-10k			
	PLCC	RMSE	RMSE*	OR	PLCC	RMSE	RMSE*	OR
Q-Align†	.940	5.251	5.166	.983	.952	.330	.155	.292
LIQE†	.915	6.234	6.149	.983	.720	.751	.526	.703
QualiCLIP	.771	9.837	9.752	.990	.673	.800	.572	.725
TOPIQ†	.978	3.198	3.124	.959	.546	.907	.671	.769
HyperIQA†	.956	4.507	4.428	.977	.503	.936	.700	.776
DBCNN†	.935	5.483	5.399	.982	.499	.939	.700	.799
MUSIQ†	.904	6.612	6.527	.985	.596	.869	.641	.730
CLIP-IQA+†	.863	7.790	7.705	.987	.665	.809	.581	.727
CLIP-IQA	.724	10.647	10.561	.992	.521	.924	.691	.787
SI	.561	12.782	12.696	.993	.328	1.023	.783	.821
SIFT+BoVW (canon.)	.484	13.508	13.423	.995	.224	1.055	.814	.835
Keypoint density	.484	13.502	13.417	.994	.227	1.055	.814	.836
NIQE	.393	14.194	14.108	.993	.439	.973	.734	.807
BRISQUE	.253	14.931	14.844	.995	.395	.995	.758	.807
ResNet-50	.162	15.231	15.143	.994	.194	1.062	.821	.834
DINOv2	.065	15.402	15.315	.996	.041	1.082	.841	.840

Table S5 Mean direction-aligned SRCC by paradigm. The two blocks are separated because the change between databases means different things in each. In the upper block no member was fitted on either database, so Δ compares two generalization results. In the lower block at least one member was fitted on KonIQ-10k, marked †; there the KonIQ-10k column mixes fit with generalization and no Δ is meaningful.

Paradigm	*N*	KADID	KonIQ	Δ	Best member
Neither database seen by any member					
ITU-T complexity	1	0.219	0.513	+0.294	SI
SIFT+BoVW*	2	0.124	0.446	+0.322	canonical IDF
Handcrafted	1	0.123	0.447	+0.324	Keypoint density
NR-IQA, NSS	2	0.355	0.302	−0.053	NIQE
CNN+BoVW	19	0.318	0.058	−0.260	VGG-19+BoVW
Deep features	2	−0.086	−0.115	−0.029	DINOv2 ViT-S/14
At least one member fitted on KonIQ-10k					
NR-IQA, LMM	1	**0.951**†	**0.920**†	—	Q-Align
NR-IQA, CNN reg.	2	0.473	**0.933**†	—	HyperIQA
NR-IQA, transformer	2	0.534	**0.923**†	—	TOPIQ
NR-IQA, CLIP	4	0.636	**0.774**†	—	LIQE

*Mean over the magnitudes of the two TF-IDF variants, which differ in orientation but not in strength. Restricted to the two CLIP members fitted on neither database, that paradigm gives 0.583 on KADID-10k and 0.679 on KonIQ-10k, a change of +0.096.

Table S6 The BoVW variants in the domain, on the agreement axis and on what they deliver. *Differs* counts the clusters in which the variant selects a different segment from the HSMVideoSumm one; *Agreement* is top-1 agreement with the reference model over the 15 clusters; r_{kp} is the rank correlation with keypoint density over the 33 segment slots; and the last three columns are the summary the variant delivers, as in Table S10. The comparison between summing and averaging defined in the main article is therefore evaluated directly. The final column is the uncalibrated raw sum of the DOVER technical and aesthetic branches, used only as a scale-sensitivity diagnostic.

Variant	Aggr.	Differs	Agreem.	r_{kp}	Dur. (s)	Q-Align	raw T+A
SIFT+BoVW (as implemented)	sum	—	9/15	+0.591	634.0	61.1%	52.1%
SIFT+BoVW (canonical IDF)	sum	1/15	10/15	+0.116	801.5	62.2%	47.6%
BoVW / descriptor count	mean	2/15	10/15	−0.199	810.9	59.0%	48.2%
BoVW / frame count	mean	1/15	10/15	−0.145	**459.3**	**77.3%**	**71.5%**
Keypoint density	mean	5/15	7/15	+1.000	—	—	—
LIQE (for reference)	mean	5/15	11/15	−0.381	455.6	85.8%	91.8%

Table S7 Descriptive in-domain agreement with the Q-Align reference model over the 15 redundancy clusters, raw and chance-corrected against $\bar{p}_e = 0.472$. The main article uses the larger 42-cluster set. These clusters share topics and two pairs share a segment; the table therefore makes no cluster-independence inference. Topic-block intervals for the larger system-defined set are reported in the main article. Roles: D = the HSMVideoSumm operation, C = candidate, X = diagnostic.

Method	Role	Agreement		$\kappa_{\mathrm{uniform}}$
MUSIQ	C	13/15	(86.7%)	+0.747
DBCNN	C	12/15	(80.0%)	+0.621
TOPIQ	C	12/15	(80.0%)	+0.621
LIQE	C	11/15	(73.3%)	+0.495
HyperIQA	C	11/15	(73.3%)	+0.495
NIQE	C	10/15	(66.7%)	+0.368
QualiCLIP	C	10/15	(66.7%)	+0.368
BRISQUE	C	10/15	(66.7%)	+0.368
SIFT+BoVW	D	9/15	(60.0%)	+0.242
CLIP-IQA	C	8/15	(53.3%)	+0.116
CLIP-IQA+	C	8/15	(53.3%)	+0.116
DINOv2	X	5/15	(33.3%)	−0.263

Table S8 The four redundancy clusters in which the method under study diverges from the majority of the quality references. Each row is a candidate segment; each column a scoring method. The top-1 choice of each method is shaded: pink for SIFT+BoVW, green for the quality references. NIQE and BRISQUE are lower-is-better. Scores are reproduced from the aggregated per-cluster results.

Segment	SIFT+BoVW	LIQE	NIQE↓	Q-Align	QualiCLIP	BRISQUE↓
bebe_real C1						
Band 0:37 (10 s)	**5448.0**	1.49	6.55	4.02	0.28	50.97
Record 0:46 (11 s)	2077.6	**1.59**	**5.57**	**4.17**	**0.30**	**41.44**
brumadinho C2						
Band 0:00 (13 s)	2888.3	**2.60**	**5.04**	**4.84**	**0.36**	**44.95**
Record 4:24 (188 s)	**5332.7**	1.13	6.23	3.74	0.27	53.57
isencao_imposto C0						
Band 2:46 (30 s)	**3732.8**	4.23	**4.67**	3.70	**0.62**	34.65
CNN 1:12 (25 s)	3010.0	**4.47**	4.74	**4.26**	0.59	**29.84**
mounjaro C0						
Band 0:00 (8 s)	559.8	4.12	7.29	**4.38**	0.56	63.19
Dom. Esp. 2:06 (16 s)	**1945.1**	3.95	5.87	3.90	0.57	36.55
Dom. Esp. 3:07 (25 s)	1784.4	3.61	5.28	3.68	0.56	42.29
Record 0:00 (14 s)	815.2	**4.93**	**4.76**	4.26	**0.63**	**35.45**

Table S9 Eligible set as a function of the generalization threshold, under both readings of "a database it was not fitted on". *Universal* requires the threshold to hold on every unseen database (the reading adopted in the main article); *existential* requires it on at least one. The sensitivity requirement is held at a within-reference mean above 0.50 and at least 15/25 types; the topic-block lower confidence limit must also exceed the in-domain floor.

Minimum ρ^*	Universal reading (adopted)	Existential reading
0.50	LIQE, QualiCLIP, CLIP-IQA+, MUSIQ, TOPIQ, CLIP-IQA	LIQE, QualiCLIP, CLIP-IQA+, MUSIQ, TOPIQ, CLIP-IQA, DBCNN, HyperIQA
0.55	LIQE, QualiCLIP, CLIP-IQA+, MUSIQ	*same eight methods*
0.60	**LIQE, QualiCLIP, CLIP-IQA+**	*same eight methods*
0.65*	**LIQE, QualiCLIP, CLIP-IQA+**	*same eight methods*
0.70	LIQE	*same eight methods*
0.75	none	LIQE, QualiCLIP, CLIP-IQA+, MUSIQ, TOPIQ, DBCNN, HyperIQA

*The value used. At the adopted generalization threshold and universal reading, the same three methods survive for sensitivity type-count cuts from 12/25 through 22/25; none survives at 23/25. Requiring only a positive floor point estimate instead of a topic-block lower limit does not change the three-method set.

Table S10 The same comparison on the 15 redundancy clusters. The two references agree closely here; their disagreement emerges on the 42-cluster analysis in the main article, which is the primary estimate. The DOVER column uses the uncalibrated raw technical-plus-aesthetic sum, not DOVER-T.

Scoring method	Changed	by raw T+A	by Q-Align	Dur. (s)	Words
SIFT+BoVW	—	52.1%	61.1%	634.0	1,613
LIQE	5/15	91.8%	85.8%	455.6	1,124
BRISQUE	6/15	84.5%	84.7%	431.0	1,073
QualiCLIP	4/15	80.9%	80.4%	439.6	1,124
NIQE	4/15	80.9%	80.4%	439.6	1,124
DINOv2	12/15	53.4%	36.7%	328.6	904

Table S10 (continued). Construct and scale sensitivity on the 42 operational clusters.

DOVER output	Estimand	SIFT	LIQE	Δ; separate-label CI	Hierarchical CI
Technical branch	Global range (%)	50.5	76.0	+25.5; [-12.1, +54.9]	[-17.2,+59.6]
Technical branch	Raw mean	−.0770	−.0701	+.0069; [−.0028, .0178]	[−.0041, .0197]
Technical branch	Mean cluster position (%)	57.1	64.4	+7.2; [−20.7, 35.1]	[−22.4, 37.5]
Technical branch	Selected maximum (%)	54.8	59.5	+4.8; [−22.2, 31.7]	[−24.4, 34.2]
Raw T+A sum	Global range (%)	47.4	77.7	+30.3; [1.1, 54.1]	[−4.1, 59.0]
Aesthetic branch	Global range (%)	45.0	73.7	+28.7; [5.3, 49.7]	[−.2, 54.6]

The branch sum is the raw technical output plus the raw aesthetic output. It is not the official ITU-standardized DOVER fused score. The technical branch is primary; the other rows diagnose construct sensitivity. **Every interval in this continuation resamples the 19 separate topic labels**, which is the sensitivity grouping: the primary source-aware block interval for the technical branch is [-13.5, +54.6], in Section 4.6 of the main text.

DOVER-T range filter	Clusters	Threshold	Global-range Δ
All clusters	42	—	+25.5
Exclude bottom 10%	37	.00397	+25.9
Exclude bottom 25%	31	.00891	+28.1

DOVER-T cluster ranges: minimum .00028, first quartile .00891, median .01686, third quartile .04118, maximum .10653; zero-range clusters: 0.

Matched-input control: at most ten identical frames and mean pooling.

Scorer	Differs from SIFT	DOVER-T range	Raw mean	Selected max
SIFT+BoVW	—	52.0%	−.0766	57.1%
LIQE	16/42	79.1%	−.0692	64.3%

LIQE–SIFT global-range Δ = +27.1 points (source-aware block CI [5.2, 46.6]; separate-label CI [5.6, 47.8]; hierarchical CI [2.4,52.5]). Raw-branch Δ = +.00736 (separate-label [.00135, .01406]; hierarchical [.00057, .01683]). Thirteen of the 16 disagreement clusters belong to the 22-cluster complete-operation change set. Relative to the published HSMVideoSumm path, matched-input SIFT changes 12 choices and matched-input LIQE changes 22; the control isolates scorer identity, not the historical operation itself.

The complete registered commit is:

`9377c07ca080ac61c7cff80e0ff6c7507b6ebaba`.

The submission archive contains `registered_prediction_commit.bundle`; the clone, bundle verification and object-inspection commands in its README make the commit and timestamp independently inspectable even before the public release.

Table S11 The eight predictions registered at commit `9377c07` before the constructed experiment, against the measurements. None is used to filter a method.

	Subject	Prediction	Measured	Outcome
P1	SIFT+BoVW	High accuracy on blur and darkening	90%, 100%	holds
P2	SIFT+BoVW	Inverted on white noise	8%	holds
P3	SIFT+BoVW	Near chance on JPEG and H.264	6% and 90%	refuted
P4	CNN+BoVW (19)	Chance on every type	44.9%, CI [41.0, 48.9]; range 30–64	refuted
P5	NIQE, BRISQUE	Accuracy falls under heavy overlay	+2.0, +2.2 points	refuted
P6	Learned methods	No interaction with overlay	−18.1 to −8.1 for three	refuted
P7	SIFT+BoVW	Design B |mean Spearman| below 0.15 (insensitivity)	−0.425	refuted
P8	Q-Align	Design A accuracy above 0.90	98.2%, CI [97.2, 99.1]	holds

Table S11 (continued). Fidelity and descriptor-count diagnostics for the constructed broadcast pairs.

Method	All 8 types (800)			Without dark/desaturation (600)		
	Correct	Ties	Accuracy	Correct	Ties	Accuracy
SIFT+BoVW	521	0	65.1%	382	0	63.7%
LIQE	796	0	99.5%	599	0	99.8%
Q-Align	786	0	98.2%	600	0	100.0%
DOVER-T	781	0	97.6%	599	0	99.8%

Type	Mean ΔD	$\Delta D > 0$	Mean ΔD by level 1–5
Blur	−797	10%	−130, −417, −913, −1188, −1339
Dark	−796	0%	−229, −487, −791, −1116, −1358
H.264	−398	10%	−128, −220, −361, −541, −737
JPEG	+289	95%	+161, +252, +339, +376, +317
Motion blur	−1067	0%	−548, −909, −1174, −1308, −1396
Noise	+469	92%	+126, +348, +517, +607, +749
Rescale	−646	12%	−154, −377, −585, −911, −1204
Desaturation	+18	61%	0, +14, +21, +34, +24

$\Delta D = D_{\text{degraded}} - D_{\text{unmodified}}$. Strict accuracy uses correct/total; a tie is neither correct nor incorrect and remains in the denominator.

DOVER-T	dark	motion	blur	H.264	rescale	satur.	noise	JPEG	All
Accuracy (%)	93	100	100	100	100	89	99	100	97.6
Mean severity ρ^*	.985	.900	1.000	1.000	1.000	.715	.995	1.000	—

The raw technical output is the exact branch later used for the 42-cluster output analysis. Its overall segment-block 95% CI is [96.1, 99.0]; after excluding darkening and desaturation, accuracy is 99.8% with CI [99.5, 100]. There are no ties. Each condition is encoded through the same 5-fps MP4 wrapper and uses paired spatial fragment seed 42.

Table S12 Method inventory and released-weight supervision. *Sup.*: OU = opinion-unaware, K = KonIQ-10k, L = LIVE, M = multi-database mixture. Roles: D = the HSMVideoSumm operation, C = candidate, R = reference, X = diagnostic or ablation. The daggered partial-coverage methods are excluded from complete-case screening.

Method	Family	Construct	Sup.	Pol.	GPU	Role
SIFT+BoVW	Handcrafted	TF-IDF word count	–	↑	No	D
Keypoint density	Handcrafted	keypoints/frame	–	↑	No	X
DINOv2	Self-supervised	patch-token norm	–	↑	Yes	X
ResNet-50	Supervised	pooled-feature norm	–	↑	Yes	X
BRISQUE	NSS	natural-statistics deviation	L	↓	No	C
NIQE	NSS	pristine-model distance	OU	↓	No	C
QualiCLIP	CLIP	quality score	OU	↑	Yes	C
LIQE	CLIP	quality, distortion, scene	K	↑	Yes	C
Q-Align	LMM	textual quality levels	M	↑	Yes	R
SI	ITU-T complexity	Sobel energy	–	↑	No	X
19 CNN+BoVW	ablation	TF-IDF word count	–	↑	Yes	X
DBCNN	CNN regression	bilinear pooling	K	↑	Yes	C
HyperIQA	CNN regression	adaptive weights	K	↑	Yes	C
MUSIQ	Transformer	multi-scale score	K	↑	Yes	C
TOPIQ	Transformer	semantic attention	K	↑	Yes	C
CLIP-IQA	CLIP	antonym prompts	OU	↑	Yes	C
CLIP-IQA+	CLIP	tuned prompts	K	↑	Yes	C
ARNIQA†	Self-supervised	distortion manifold	K	↑	Yes	partial
TReS†	CNN–transformer	ranking/consistency	K	↑	Yes	partial

PyIQA registry keys used by the archived scorers are `brisque`, `niqe`, `dbcnn`, `hyperiqa`, `musiq`, `topiq_nr`, `clipiqa`, `clipiqa+`, `liqe`, `qualiclip`, `arniqa` and `tres`. The final inventory and analysis archive were frozen on 23 August 2026. MANIQA was attempted with PyIQA 0.1.15.post2 and the `maniqa-kadid` checkpoint. A single-image diagnostic completed, but the released implementation evaluates 20 crops per image and the full 21,360-image three-database run did not finish within the experiment window on the single 8-GB RTX 4060; it is therefore an uncompleted method, not a failed candidate.

Table S12 (continued). Complete extractor inventory for the CNN+BoVW ablation. Every row uses the final spatial tensor before global pooling or classification.

Extractor	Released torchvision weights	Spatial endpoint
ResNet-18	IMAGENET1K_V1	trunk before average pooling
ResNet-50	IMAGENET1K_V2	trunk before average pooling
ResNet-101	IMAGENET1K_V2	trunk before average pooling
VGG-16	IMAGENET1K_V1	full `features` module
VGG-19	IMAGENET1K_V1	full `features` module
EfficientNet-B0	IMAGENET1K_V1	full `features` module
MobileNet-V3-Large	IMAGENET1K_V2	full `features` module
ConvNeXt-Tiny	IMAGENET1K_V1	full `features` module
ConvNeXt-Small	IMAGENET1K_V1	full `features` module
DenseNet-121	IMAGENET1K_V1	full `features` module
DenseNet-201	IMAGENET1K_V1	full `features` module
ResNeXt-50-32x4d	IMAGENET1K_V2	trunk before average pooling
ResNeXt-101-64x4d	IMAGENET1K_V1	trunk before average pooling
Wide-ResNet-50-2	IMAGENET1K_V2	trunk before average pooling
Wide-ResNet-101-2	IMAGENET1K_V2	trunk before average pooling
Inception-V3	IMAGENET1K_V1	convolutional trunk through Mixed_7c
EfficientNet-B4	IMAGENET1K_V1	full `features` module
SqueezeNet-1.1	IMAGENET1K_V1	full `features` module
MobileNet-V2	IMAGENET1K_V2	full `features` module

Common preprocessing: bicubic resize to 256 pixels, center crop to 224×224, ImageNet mean and standard deviation, and FP16 inference. Each spatial location is flattened into one descriptor and ℓ_2-normalized in FP32. The same internal key-frame filter, vocabulary construction, TF–IDF weighting and segment aggregation are then applied. The archived implementation is `cnn_bovw_scorer.py`. **SURF is not among the extractors**: it is patented and its OpenCV module ships only in a non-free build compiled with `OPENCV_ENABLE_NONFREE`, so `cv2.xfeatures2d` is absent from the OpenCV 4.13.0 installation used here, while SIFT, ORB, AKAZE and BRISK are all present.

Table S12 (continued). Exact configurations of the two model-based references used in the broadcast analyses.

Model	Config.	Input / pooling	Output used
Q-Align	`q-future/one-align`; task `quality`, input `image`	Uniform 1-fps frames; arithmetic mean of frame scores	Four-bit NF4 with double quantisation and FP16 compute; higher score is better. The same configuration scores the 800 constructed pairs and supplies the in-domain floor.
DOVER	Official `DOVER.pth` and `dover.yml`, `val-l1080p` view decomposition	Technical: 7×7 spatial fragments of size 32, 32 frames, interval 2, three clips; aesthetic: 224×224, 32 frames, interval 2, one clip	The technical branch is the primary output reference and is also evaluated on the 800 pairs with a common 5-fps MP4 wrapper and paired seed 42. The raw branch sum is only a sensitivity analysis; it is not the official standardised fusion or DOVER++.

Table S13 Computational cost measured on one RTX 4060 over the same inputs. Frame-wise entries report median latency after ten warm-up and sixty timed iterations; the final two rows are measured complete paths.

Method	ms/frame	IQR	Per cluster
SI	2.2	0.14	0.2 s
QualiCLIP	14.3	1.68	1.2 s
TOPIQ	18.7	0.36	1.5 s
BRISQUE	19.6	0.98	1.6 s
CLIP-IQA+	21.4	0.65	1.8 s
DBCNN	21.5	0.46	1.8 s
LIQE	22.7	0.82	1.9 s
CLIP-IQA	25.2	0.51	2.1 s
NIQE	34.6	0.90	2.8 s
MUSIQ	40.0	9.36	3.3 s
ResNet-50	46.2	8.44	3.8 s
DINOv2	46.4	36.63	3.8 s
Keypoint density	46.8	1.48	3.8 s
HyperIQA	50.1	0.77	4.1 s
SIFT+BoVW	$\sim$ 1,724	—	182 s
Q-Align	322.6	—	26 s

The uniform 1-fps candidate cache contains 2,892 frames from 83 distinct segments. Counting repeated memberships across the 93 cluster slots gives 3,525 segment-frame instances. Candidate latency was timed per decoded frame; the SIFT+BoVW row is the complete operation divided by its input-frame count and therefore includes key-frame selection, descriptor extraction, vocabulary quantisation and aggregation. The per-cluster column is the operational quantity; the per-frame SIFT value is an approximate normalization rather than an identical-path microbenchmark.

Table S14 Descriptive three-database contrasts and released-weight exposure. K–C is KonIQ-10k minus CLIVE; C–KADID is CLIVE minus KADID-10k. Positive values are losses. K = KonIQ-10k opinion training, OU = opinion-unaware and L = LIVE opinion training. The contrasts do not isolate causal effects because the databases, architectures and objectives are unmatched.

Method	Sup.	KonIQ	CLIVE	KADID	K–C	C–KADID
Released checkpoint fitted on KonIQ-10k						
TOPIQ	K	0.973†	0.811	0.511	+0.162	+0.300
HyperIQA	K	0.947†	0.750	0.462	+0.197	+0.288
DBCNN	K	0.919†	0.756	0.484	+0.163	+0.272
MUSIQ	K	0.873†	0.789	0.556	+0.084	+0.233
CLIP-IQA+	K	0.824†	0.805	0.654	+0.019	+0.151
LIQE	K	0.915†	0.865	0.723	+0.050	+0.142
mean over the six					+0.113	+0.231
Never fitted on KonIQ-10k						
QualiCLIP	OU	0.711	0.757	0.665	−0.046	+0.092
CLIP-IQA	OU	0.647	0.703	0.501	−0.056	+0.202
NIQE	OU	0.368	0.449	0.379	−0.081	+0.070
BRISQUE	L	0.236	0.313	0.330	−0.077	−0.017
mean over the four					−0.065	+0.087

†In-distribution. CLIVE contains 1,162 images and is unseen by every checkpoint in the table. The operational partition assigns 33% of the mean KonIQ-to-KADID gap of the fitted group to K–C and 67% to C–KADID; these percentages describe contrasts and are not causal estimates.

Table S15 Spearman association with human scores before and after conditioning on detectable keypoint density. r_{kp} is the rank correlation between the method score and keypoint density. Differences are descriptive attenuations, not causal fractions of "content density."

Method	KonIQ-10k				KADID-10k			
	r_{kp}	$\lvert\rho\rvert$	$\lvert\rho_{\cdot\mathrm{kp}}\rvert$	Diff.	r_{kp}	$\lvert\rho\rvert$	$\lvert\rho_{\cdot\mathrm{kp}}\rvert$	Diff.
Q-Align	+.44	.920	.901	−.019	+.13	.951	.950	−.001
LIQE	+.43	.915	.894	−.020	+.18	.723	.718	−.005
TOPIQ	+.48	.973	.966	−.007	+.21	.511	.500	−.011
HyperIQA	+.48	.947	.933	−.014	+.17	.462	.451	−.011
DBCNN	+.49	.919	.898	−.021	+.22	.484	.472	−.012
MUSIQ	+.44	.873	.841	−.032	+.20	.556	.546	−.010
QualiCLIP	+.53	.711	.626	−.086	+.29	.665	.663	−.002
CLIP-IQA	+.27	.647	.611	−.036	+.22	.501	.489	−.012
SI	+.77	.513	.296	−.218	+.70	.219	.187	−.032
NIQE	−.63	.368	.125	−.243	−.33	.379	.362	−.018
BRISQUE	−.31	.236	.116	−.120	−.13	.330	.319	−.011
SIFT+BoVW (canonical)	+.998	.447	.016	arith.	+.992	.125	.028	arith.
SIFT+BoVW	−.9999	.445	n.d.	arith.	−.9998	.123	n.d.	arith.

The shaded rows are the near-tautological regime: conditioning a score on a variable with which it is almost perfectly correlated removes the variable from itself. For the HSMVideoSumm row the partial coefficient is not reported because the denominator approaches zero and the result is numerically unstable.

Table S16 The sixteen operations audited, each linked to the objective its source states and the role it has there. *Tested reading* is the one this article evaluates: higher score means better technical quality, applied to a single frame without a model. No source is claimed to be wrong about its own objective; what is tested is the reuse. *Inverts* counts degradation types on which the operation ranks the degraded copy higher (Table S27). Where two works adopt the same operation the use can differ: Laplacian variance rejects low-light frames alongside grey-scale intensity and a Canny detector in Ref. [12], and enters $0.7\,z(\text{sharp}) + 0.3\,z(\text{bright})$ in Ref. [44].

Operation	Objective in the source	Role there	Inverts
Averaged aesthetics (3) [24]	Keyframe interestingness	three aesthetic terms; the source gives no combination, and the mean is built here	1/8
Colourfulness (Hasler) [25]	Perceived colourfulness of a scene	stand-alone perceptual index	1/8
Colourfulness (Ke) [42]	Professional vs. snapshot photograph	feature of a naive Bayes classifier	1/8
Composite of five [78]	Keyframe ranking	unweighted mean of five features	1/8
Contrast [42]	Professional vs. snapshot photograph	feature of a naive Bayes classifier	2/8
Edge distribution [42]	Professional vs. snapshot photograph	feature of a naive Bayes classifier	2/8
Entropy [70]	Keyframe diversity and coverage	SE in segment coverage (Eq. 21)	2/8
GLCM contrast [90]	Keyframe extraction	texture descriptor	1/8
Gradient magnitude [70]	Keyframe diversity and coverage	GM in visual attention (Eq. 20)	3/8
Gradient orientation [70]	Keyframe diversity and coverage	GO in segment diversity (Eq. 22)	4/8
Grey-level dispersion [34]	Static-frame selection	one of four inputs to a cost multiplicative in cluster distance	2/8
Hue count [42]	Professional vs. snapshot photograph	feature of a naive Bayes classifier	5/8
Laplacian variance [12]	Blur rejection in frame filtering	sharpness threshold	1/8
Saliency area [70]	Keyframe diversity and coverage	VS in visual attention (Eq. 20)	6/8
Saturation [29]	Aesthetic scoring	one term among several	3/8
Spatial information [31]	Spatial detail of a test sequence	characterization, not quality	—

Tested reading, identical for the fifteen with a declared direction: the operation is computed on one frame and read as a technical-quality score in which a higher value marks the better of two versions of the same content. Fifteen are closed-form; saliency area requires a learned map in its source and is marked as such.

Table S17 The sixteen visual-selection operations the video-summarization literature uses at the selection stage, reimplemented from the published equations and measured under one protocol. *Sensitivity* is the SRCC against the degradation level within each KADID-10k reference, aligned to the direction the source declares for its own statistic, averaged over the 25 distortion types, with the count of types above .50; *Generalization* is the lowest such correlation on any database the method was not fitted on; *Floor* is chance-corrected agreement on the operational clusters. **The two families use different reference models there**: operations against LIQE, the ten no-reference models against Q-Align, so that column is not a comparison between families. The damage columns share one protocol. The rule admits a method when sensitivity exceeds .5 on at least 15 of 25 types and generalization exceeds .65. **No operation clears either external screening criterion**: the best sensitivity is .298 and the best generalization among the fifteen evaluated is .139. Spatial information declares no direction, carries no verdict and is excluded from every count. **The fifteen are not fifteen independent measurements**: on KADID-10k their score matrix has rank thirteen, because one colourfulness implementation duplicates another exactly and one operation is the mean of three others. The *Floor* column reports the in-domain agreement against the reference model with the lower end of its 95% interval in brackets, so a pass can be read for how much of it survives the interval: the weakest passing operation in this table has a lower bound of +.016.

Operation	Source	Sens.	Types	Gen.	Floor κ [95% lo]	Elig.
Edge distribution	[42]	.298	9/25	+.083	−.124 [−.388]	no
Gradient orientation	[70]	.291	12/25	−.035	+.461 [+.200]	no
Spatial information	[31]	.251	13/25	+.218	+.506 [+.279]	—
Contrast	[42]	.185	10/25	+.139	+.326 [+.056]	no
Grey-level dispersion	[34]	.149	11/25	+.096	+.191 [−.071]	no
Hue count	[42]	.147	6/25	−.162	−.258 [−.494]	no
Entropy	[70]	.117	11/25	+.102	+.236 [−.024]	no
Gradient magnitude	[70]	.104	11/25	+.129	+.551 [+.380]	no
Saliency area	[70]	.075	3/25	+.031	+.101 [−.196]	no
Averaged aesthetics (3)	[24]	.032	8/25	+.027	+.461 [+.108]	no
Composite of five	[78]	.029	11/25	−.020	+.371 [+.094]	no
GLCM contrast	[90]	.019	9/25	+.136	+.461 [+.247]	no
Laplacian variance	[12]	−.047	9/25	+.095	+.506 [+.284]	no
Saturation	[29]	−.092	7/25	−.254	+.011 [−.311]	no
Colourfulness (Hasler)	[25]	−.182	4/25	−.062	+.281 [+.016]	no
Colourfulness (Ke)	[42]	−.182	4/25	−.062	+.281 [+.016]	no

Table S18 Which operation entered which test. A filled marker means the operation has a result on that axis. **Every absence in this table is a property of the source, not of our sequence of work.** Spatial information carries no paired verdict and no downstream choice because its source declares no direction, so selecting the larger value would be arbitrary; it is the only absence, and it is the same reason in both columns. **The in-domain floor is computed on a single testbed**: every operation is scored on the same frames as the reference model, which was not the case in an earlier version of this column, where eleven operations were sampled by one decoder and the reference by another. Recomputing the column with the decoders matched moves no verdict, and the largest change is three clusters of agreement; the operations that move are those whose top two candidates differ by less than the median margin, so the effect is tie-breaking at the discretisation step of the agreement statistic rather than a change in what the operation measures. Sensitivity and generalization cover all 16. Counts per axis are in the last row.

Operation	Paired	Sens.	Gen.	Floor	Down.	Absent because
Averaged aesthetics (3)	●	●	●	●	●	
Colourfulness (Hasler)	●	●	●	●	●	
Colourfulness (Ke)	●	●	●	●	●	
Composite of five	●	●	●	●	●	
Contrast	●	●	●	●	●	
Edge distribution	●	●	●	●	●	
Entropy	●	●	●	●	●	
GLCM contrast	●	●	●	●	●	
Gradient magnitude	●	●	●	●	●	
Gradient orientation	●	●	●	●	●	
Grey-level dispersion	●	●	●	●	●	
Hue count	●	●	●	●	●	
Laplacian variance	●	●	●	●	●	
Saliency area	●	●	●	●	●	
Saturation	●	●	●	●	●	
Spatial information	○	●	●	●	○	source declares no direction
of 16	**15**	**16**	**16**	**16**	**15**	

Dependence among the audited operations.

Two identities inside the inventory make some of the operations redundant by construction: the two colourfulness implementations are the same Hasler expression, and one further operation is the unweighted mean of three others. Counted over the fifteen audited operations, the raw score matrix has rank thirteen, with an effective 8 independent directions. **This is a statement about dependence, not about the tested family**: the Holm correction of the main article is applied to the 120 operation–type comparisons actually made, and is not relaxed to 8.

The aggregation the counting instance uses.

Retained descriptors are quantized into a vocabulary of $K = 300$ visual words and the segment score is the sum $S_{\mathrm{BoVW}}(s) = \sum_i \sum_{v=1}^{K} \mathrm{tf}(v, f_i)\,\mathrm{idf}(v)$ over the frames its own key-frame selector retains, rather than the arithmetic mean over uniformly sampled frames that every still-image scorer receives.

The two inverse-document-frequency formulations, and what separates them.

The inverse document frequency in the counting instance the testbed ships with uses the vocabulary size, $\mathrm{idf}_{\mathrm{dep}}(v) = \log_{10}(K/\mathrm{df}(v))$, whereas canonical TF-IDF uses the number of documents, $\mathrm{idf}_{\mathrm{can}}(v) = \log_{10}(N/\mathrm{df}(v))$. Their relation is exact, $S_{\mathrm{dep}}(s) = S_{\mathrm{can}}(s) + D_s \log_{10}(K/N)$, where D_s is the number of descriptors, so the two differ by an *additive count term* and not by rescaling. The identity holds within 3×10^{-11} on all 93 segment slots. At cluster scale $N \in \{2, 3, 4\}$, so the additional term rewards count and contributes a median 98.3% of the score; on a database of approximately 10^4 documents the term is negative and reverses the count ordering.

The count channel against an external association.

On KonIQ-10k the coefficient of variation is .938 for D_i and .098 for the mean IDF weight, and the canonical score correlates .998 with descriptor count. It shows the same MOS/DMOS association as keypoint density, within the operational $\pm.02$ margin: $\rho^* = .447$ for both, with score-to-keypoint correlation .998. Steiger's $Z = .199$ is below the Bonferroni-corrected critical value of 3.95, and both one-sided equivalence tests reject. KADID gives the same association-level result under reference-block resampling, .125 for canonical BoVW against .123 for keypoint density, paired difference $+.0024$, 95% CI $[-.0019, +.0084]$. The HSMVideoSumm formulation is almost exactly the sign-reversed count on the large databases ($-.9999$ on KonIQ, $-.9998$ on KADID and $-.9947$ on CLIVE). Within segment the score increases with severity for noise ($\rho = +.720$, CI $[+.455, +.925]$) and JPEG ($+.640$, CI $[+.395, +.850]$), and the mean noise count rises at every severity level.

Exchanging the extractor, and why normalization does not repair the reading.

Across the 19 convolutional substitutions the mean $|\rho|$ is .090 on KonIQ and .112 on CLIVE, with no value above .238 on either database; KADID ranges from .156 to .418. Fixed spatial feature maps remove per-image count variation on the image databases, leaving the rarity channel measured by the ablation. On broadcast segments the count varies again, and **what carries that variation is detail per frame rather than the number of frames the selector keeps**: keypoints per frame track the score at $+.599$, the frame count only at $+.177$, and replacing the score by the frame count preserves 23 of 42 representative choices against 42 for the descriptor count (Table S33). **Simple normalization does not repair the mechanism.** Dividing by descriptor count exposes the weak mean-IDF-weight channel already measured by the ablation; dividing by frame count is mismatched to that operation because its internal key-frame selector makes descriptor count nearly independent of duration. On

the 42 clusters, frame normalization changes the uncalibrated DOVER two-branch-sum output only from 47.4% to 50.6% and induces an inverse duration association ($\rho = -.52$). The per-scorer delivered outputs behind these two values, with their intervals, are in the reproducibility archive.

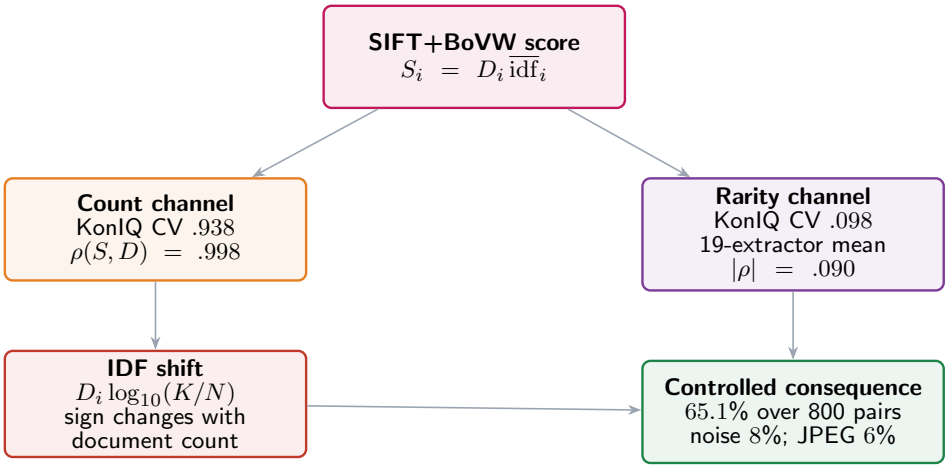


Fig. S1 Measured mechanism. The factorization identifies two channels; the data determine their relative variation and connect count dominance to the controlled-degradation failures.

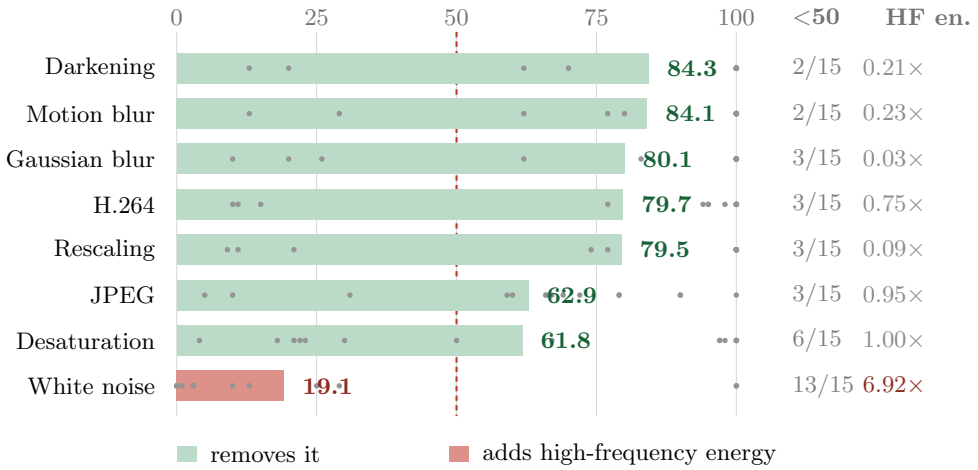


Fig. S2 Mean paired accuracy of the fifteen summarization operations, by degradation type; grey dots are the individual operations and the dashed line is chance. **Rows are ordered by mean accuracy, descending.** The last column is what each degradation does to local image high-frequency energy, measured as the median ratio to the original. For the Seven transformations that reduce it, mean paired accuracy exceeds 50 per cent and the displayed median ratio is at most 1.00. **White noise is the only one that raises it**, by a factor of 6.9, and it is answered at 19.1 per cent with a median of 3.0. The middle column counts operations below chance.

Table S19 Model-reference output over the 42 operational redundancy clusters. Changed counts representatives differing from the SIFT+BoVW operation. DOVER-T is the raw DOVER technical branch and is not used in screening; Q-Align defines the in-domain screening floor. Both outputs use the global-achievable-range estimand of Eq. 2 of the main article; Max is the percentage of clusters in which the selected representative has the highest DOVER-T score.

Quality operation	Changed	DOVER-T	Max	Q-Align
SIFT+BoVW	—	50.5%	54.8%	48.6%
LIQE	**22/42**	**76.0%**	59.5%	90.2%
QualiCLIP	19/42	71.9%	59.5%	80.6%
CLIP-IQA+	22/42	70.7%	50.0%	74.3%

Table S20 Complete-case method screening. Sensitivity reports the within-reference mean and number of KADID types above .50. Generalization is the lowest signed ρ^* over all unseen databases. Floor is normalized agreement with Q-Align on the common 42 clusters and its lower bound (L) under the 19 separate topic labels, **which is the sensitivity grouping**: the primary interval of Section 4.5 of the main text resamples the 18 source-aware blocks and gives [.273, .792] for LIQE Sup.: K = KonIQ-10k opinion training; L = LIVE; OU = opinion-unaware; – = no opinion training; M = multi-database mixture. A dagger marks a value measured on a database the method trained on: Q-Align's mixture includes KADID-10k. Q-Align is a reference, not a candidate.

Method	Sup.	Pol.	Sensitivity	Lowest unseen ρ^*	Floor; L	Eligible
LIQE	K	↑	.807; 22	.723	.551; .284	**yes**
QualiCLIP	OU	↑	.849; 22	.665	.371; .143	**yes**
CLIP-IQA+	K	↑	.808; 22	.654	.236; .034	**yes**
TOPIQ	K	↑	.721; 20	.511	.506; .289	no
MUSIQ	K	↑	.733; 20	.556	.551; .297	no
DBCNN	K	↑	.668; 19	.484	.461; .274	no
HyperIQA	K	↑	.673; 19	.462	.371; .122	no
NIQE	OU	↓	.607; 19	.368	.281; .012	no
BRISQUE	L	↓	.657; 19	.236	.371; .136	no
CLIP-IQA	OU	↑	.645; 20	.501	.371; .045	no
SIFT+BoVW	–	↑	−.077; 5	−.445	.056; −.308	no
Q-Align	M	↑	.880†; 23†	–	–	reference

Table S21 Two published expressions against the property the same paragraph claims for them. **Above**: the contrast term of the Qaroush visual-attention objective, over a sweep in which contrast rises by construction (25 levels, seed 42). Histogram equalisation raises the standard deviation of every frame in this sweep. Over it, the published form and the form without the absolute value are both perfectly inversely associated with the measure used to build the inputs; only the ratio, which the equation does not write, is perfectly positively associated. **Below**: the brightness term of the Korolkov module, over a sweep in which exposure rises by construction (21 levels). The source states that it "penalizes under- and over-exposed frames", which requires a non-monotone function; the mean luminance it specifies is monotone and enters the module with weight +0.3, so a brighter frame always scores higher. **No corpus is involved.** The Korolkov inversion follows from the expression; the Qaroush result is the behaviour of the expression over this sweep.

Reading	Expression	ρ with the property
Qaroush contrast, against contrast (Eq. 8)		
Eq. 8 as published	$\frac{1}{K}\sum_k \lvert\sigma_p(\mathrm{HE}(f_k)) - \sigma_p(f_k)\rvert$	−1.0000
without the absolute value	$\frac{1}{K}\sum_k \left(\sigma_p(\mathrm{HE}(f_k)) - \sigma_p(f_k)\right)$	−1.0000
ratio, unwritten in the source	$\frac{1}{K}\sum_k \sigma_p(f_k)/\sigma_p(\mathrm{HE}(f_k))$	+1.0000
Korolkov brightness, against exposure		
as published	$\frac{1}{\lvert I\rvert}\sum_{x,y} L(I)_{x,y}$ (LAB)	+1.0000
illustrative non-monotone penalty	$-\lvert L - 127.5\rvert$	−.4883

Table S22 Traceability of every audited operation: the source work, the formula as the source states it, the function that computes it, and the reference model used on the in-domain column. Every operation name and count printed elsewhere in this article is derived from this table. **Dir.** is the direction the source declares; spatial information declares none, which is why it is measured but carries no verdict. **A superscript marks fidelity**: [a] adapted input, [b] expression not stated by the source, [c] combination chosen here, [d] adapted normalization; unmarked is the source expression, unadapted. **A marked row characterizes that variant, not the complete source system.**

Operation	Source	Formula, as the source states it	Function in code	Dir.	Ref.
Averaged aesthetics (3)[c]	[24]	three aesthetic attributes; unweighted mean[‡]	`gygli_media_crua`	↑	LIQE
Colourfulness (Hasler)	[25]	$M^{(3)} = \sigma_{rgyb} + .3\mu_{rgyb}$	`hasler_colorfulness`	↑	LIQE
Colourfulness (Ke)[b]	[42]	Hasler $M^{(3)}$; Ke gives no formula	`hasler_colorfulness`	↑	LIQE
Composite of five[d]	[78], [42]	unweighted mean of the five Ke terms	`srinivas_quality`	↑	LIQE
Contrast	[42]	width of central 98% of the grey histogram (Ke §4.5)	`ke_contraste`	↑	LIQE
Edge distribution[a]	[42]	$q_a = 1 - w_x w_y$ (Ke §4.1)[¶]	`ke_edge_distribution`	↑	LIQE
Entropy	[70]	Shannon entropy over HSV histogram bins (Eq. 5)	`qaroush_entropia`	↑	LIQE
GLCM contrast	[90]	GLCM contrast descriptor	`glcm_contraste`	↑	LIQE
Gradient magnitude	[70]	mean Sobel gradient magnitude (Eq. 10)	`qaroush_gm`	↑	LIQE
Gradient orientation	[70]	SD of Sobel gradient orientations (Eq. 15)	`qaroush_go`	↑	LIQE
Grey-level dispersion[§]	[34], [89]	grey-level standard deviation	`contraste_variancia`	↑	LIQE
Hue count	[42]	$q_h = 20 - \lvert N \rvert$, $\alpha = .05$ (Ke §4.3)	`ke_hue_count`	↑	LIQE
Laplacian variance	[12], [44]	variance of the grey-scale Laplacian	`laplacian_var`	↑	LIQE
Saliency area[a]	[70]	area fraction above a saliency threshold (Eq. 6)[§§]	`qaroush_saliencia`	↑	LIQE
Saturation	[29]	mean HSV saturation (§III-A-1)	`huang_saturacao`	↑	LIQE
Spatial information	[31]	Sobel energy, ITU-T P.910	`si`	none	LIQE

Notes to Table S22. All sixteen are computed by `ops_da_literatura.py` on the same frames and pooled identically. **Ref.** is the model each operation is compared against on the in-domain column; the ten no-reference models in the main screening table use Q-Align there instead, so that column compares each family with its own reference. Colourfulness has no formula in Ke, who cites Datta; the Hasler expression is used as an explicitly documented substitution. Scores for fifteen operations are in `corpus_inteiro.json`; spatial information is scored on the extended 83-segment design by `si_no_corpus_de_video`, in `designA_estendido_83.json`.

‡ Ref. [24] lists the attributes but not their combination; **the unweighted mean is a choice made here**. §§ Spectral-residual map [27], not the learned backbone the source prescribes. ¶ Ke builds the edge map with a Laplacian per RGB channel at 100 × 100; here it is Sobel magnitude on the grey frame. § Ref. [34] states *variance of pixels* and Ref. [89] *the mean difference and standard deviation of the grayscale values*; **only the standard deviation is implemented**.

Implementation controls. With the source edge map of Ke, sensitivity goes 0.298→0.272 and types 9→7 (the two versions correlate 0.876); no verdict changes. **Within a frame the variance and the standard deviation order alike; a mean over frames does not**: under the mean of variances 7 of the 800 pairs move and no per-type verdict does.

	Darken	Motion	Blur	H.264	Rescale	Satur.	Noise	JPEG
GLCM contrast	100	100	100	100	100	67	**0**	72
Laplacian variance	100	100	100	95	100	50	**0**	66
Entropy	100	77	83	**10**	77	97	**0**	66
Gradient magnitude	100	100	100	100	100	**23**	**0**	**5**
Gradient orientation	100	80	**26**	95	**9**	**21**	**0**	100
Saturation	**13**	100	100	**15**	100	100	**1**	90
Averaged aesthetics (3)	100	100	100	100	100	100	**3**	67
Colourfulness (Hasler)	100	100	100	100	100	100	**3**	79
Colourfulness (Ke)	100	100	100	100	100	100	**3**	79
Hue count	**20**	**13**	**10**	77	**11**	**4**	100	59
Contrast	100	100	100	98	100	**30**	**10**	60
Saliency area	62	**29**	**20**	94	**21**	**18**	**13**	**10**
Edge distribution	70	62	62	**11**	74	97	100	**31**
Grey-level dispersion	100	100	100	100	100	**22**	**25**	90
Composite of five	100	100	100	100	100	98	**29**	69

strict accuracy; red marks $L > W$

Fig. S3 Paired accuracy of each summarization heuristic on 800 unmodified–degraded broadcast pairs. Each cell is 100 pairs, 20 segments at five severity levels, and the better member is fixed by construction. **Every operation falls below chance on at least one type, and thirteen of the fifteen do so under white noise**, five of them on every pair. Rows are ordered by worst cell.

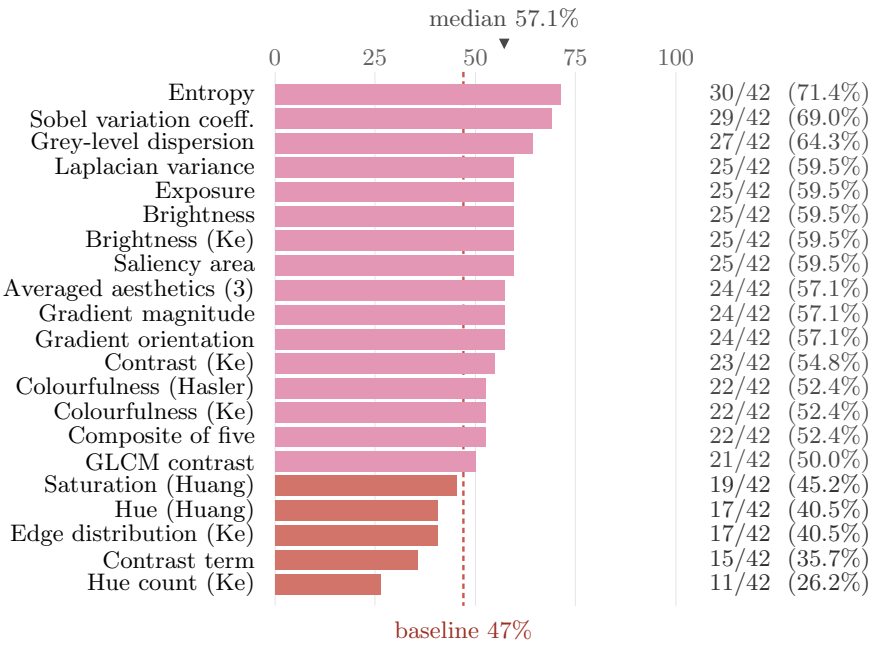


Fig. S4 Selected-segment retention under alternative implementations of the quality criterion. Only the scoring implementation is replaced; the bar is the percentage of the 42 redundancy-cluster representatives that survive the swap. **The median operation keeps 57.1 per cent of the decisions against a uniform-choice baseline of 47**, and the five in red keep fewer than chance. Retention ranges from 26 to 71 per cent. The clusters are shared across all twenty-one operations and the SIFT+BoVW one; **the frames are not**: the per-frame operations score one frame per second, while SIFT+BoVW runs its own internal key-frame selector.

Table S23 Origins, documented selection roles and evaluated implementations of the audited operations. Where origin and use fall in the same work the middle column says so, and the last column marks our own adaptations as such. The full inventory, with formulas and results files, is in Tables S16 and S22.

Operation	Origin and role there	Documented use in selection	Evaluated here
Edge distribution, hue count, contrast	Ke et al. [42]: features of a naive Bayes classifier for photo quality	Srinivas et al. [78] cite Ke for all three (§3.1.3–3.1.5) and average them into a per-frame quality score	each term alone, plus the mean and an adapted classifier
Colourfulness in that same score	Swain and Ballard, by histogram intersection, as Srinivas et al. [78] state	Srinivas et al. [78], §3.1.1	the Hasler expression instead, a declared substitution
Sharpness and brightness	Korolkov [44]: two terms of a keyframe score	the same work, $0.7\,z(\text{sharp}) + 0.3\,z(\text{bright})$	the published rule, with our normalization population stated
Grey-level dispersion	Jeong et al. [34]: a badness term inside a cost multiplicative in cluster distance	the same work, within that objective	the isolated statistic, read in the increasing direction
Gradient magnitude, saliency, entropy, gradient orientation	Qaroush et al. [70]: terms of four numbered objectives	the same work; two of the four objectives are not about technical quality	each term alone, and the visual-attention module of Eq. 20

Table S24 What each source work reports as evidence for its own visual criterion. Read from the full text of each paper; the excerpt is verbatim. *Tests* names what the source puts to a test, which is not always the operation the later literature borrows. One of the eleven source works is not in our archive and carries no verdict here: **absence of a reading is not absence of validation**. It documents the clause carried on each band of the audit table in the main text.

Source	Tests	What it reports	Verbatim excerpt
Ke et al. [42]	**human judgement of the property**	community ratings on 60,000 photos as ground truth; per-feature precision–recall and a combined classifier at 27.8 per cent equal error, on the extremes only	*"the photos have been rated by its community of users, giving us ground truth on the quality of each photo . . . The top and bottom 10% of the photos were extracted . . . we chose to ignore the middle 80%"*
Hasler and Süsstrunk [25]	**human judgement of the property**	a psychophysical scaling experiment, seven categories, the expression fitted to the ratings at 94.0 per cent; the property is colourfulness	*"We set up a psychophysical category scaling experiment, and ask people to rate images using 7 categories of colourfulness. We then fit a metric to the results"*
Srinivas et al. [78]	the summary only	precision, recall and F of the selected keyframes against the ground-truth keyframes of seven open-video.org videos	*"results are compared with the ground truth keyframes provided by open-video.org"*
Qaroush et al. [70]	the summary only	precision, recall and F1 of the extracted segments against reference summaries	*"The evaluation focuses on comparing segments extracted by our system with those identified in ground truth summaries"*
Korolkov [44]	**neither the rule nor its weights**	weights set, not fitted; ablations vary segmentation parameters, and keyframes are judged by coverage and a qualitative review	*"We set wsharp = 0.7, wbright = 0.3 to favor focused frames"*
Cakmak et al. [12]	the summary, with an ablation	TVSum20 and SumMe against human importance scores, with an ablation removing each modality	*"Table 1 presents the results of an ablation study analyzing the contribution of each modality. Removing any single component"*

Table S25 Evidence reported by each source work, continued. Columns as in the preceding table.

Source	Tests	What it reports	Verbatim excerpt
Yang et al. [90]	the summary, with an ablation	structural and feature ablations scored by F on SumMe over 15 annotations	*"we conducted experiments on the SumMe dataset for structural ablation of the model and for feature ablation"*
Gygli et al. [24]	the summary, with an ablation	f-measure against multiple human-created summaries per video, collected in a study with perception psychologists, plus an ablation measuring the gain of adding each feature to the set	*"We evaluate our method using multiple 'ground truth' summaries per video, which we acquired in a study in collaboration with perception psychologists"*
Jeong et al. [34]	the summary, with an optimisation ablation	summary-level human evaluation and precision/recall against labelled events, with a comparison against an optimisation-removed variant; the component-level claim here concerns the isolated grey-level statistic, which is not tested alone	*"and (5) our method without the optimization procedure"* (Section 5, Experimental results)
Huang et al. [29]	the summary only	Spearman and Kendall rank correlation against human annotations on four datasets, in canonical, augmentation and transfer setups	*"we conduct an extensive evaluation of the models we developed by utilizing four distinct datasets: SumMe, TVSum, OVP, and YouTube ... we carry out our experiments using the dataset in three different setups: canonical (C), augmentation (A), and transfer (T)"* (Section IV.A, Experiments setup)
Xu et al. [89]	the summary, with an ablation	F against summaries from five users on OVP, with an ablation of the frame-block length; no isolated test of the entities or of their weights	*"All the scores are normalized to the interval [0, 1] and the higher value means the better"* (Section "Scoring the entities")

Table S26 Resampling unit per analysis, and the rival unit tested against it. Observations are nested, so the unit decides the width of every interval: the italic line under each row states the nesting, and **the declared unit follows from that nesting**, observations sharing content or a source entering the same block. Each interval is also recomputed under the rival unit, and the last column reports the outcome of that recomputation.

Block	Analysis	Declared unit	Rival tested	Outcome
Block 01	Paired difference, 800 pairs	segment block	the pair	**the two units give different widths; under the declared unit the interval is** $[+29.8, +38.9]$, and $[+30.4, +39.4]$ under the coarser source-aware block (12 blocks)
	20 segments, 40 pairs each; pairs within a segment share content			
Block 01	Strict accuracy, 800 pairs	topic block	the segment block	11 of the 11 interval bounds reproduce under the declared unit, 2 under the rival
	the 20 segments come from 13 topic labels, 12 source-aware blocks after merging the two that share source videos			
Block 04	In-domain floor, 42 clusters	topic block	the cluster	3 of the 3 interval bounds reproduce under the declared unit, 1 under the rival
	the 42 clusters come from 19 topic labels, 18 source-aware blocks			
Block 04	Output contrast, 42 clusters	topic block	the cluster	over eight seeds the bounds span $[-13.3, -11.2]$ and $[54.6, 55.5]$ under the declared unit, against $[-6.9, -5.8]$ and $[52.3, 52.9]$ under the rival
	the 42 clusters come from 19 topic labels, 18 source-aware blocks			
Block 04	Regime contrast, 42 clusters	topic block	the cluster (exact McNemar)	the verdict holds under the declared unit ($p = .008$ against $p = .006$)
	the 12 discordant pairs fall in 9 topics; the null enumerates all 2^9 *block sign assignments*			

Table S27 Every operation–type cell below the 50 per cent chance floor, with its *decisive* accuracy $W/(W + L)$, which leaves ties out of the denominator and therefore differs from the strict accuracy of the main tables wherever an operation ties. Significance is an exact sign-flip randomization with Holm across the 120 operation–type tests, reported for both nested units. Over the 20 broadcast segments, 22 of the 35 cells survive and † marks the 13 that do not; over the 12 topic blocks, the more conservative unit, 11 survive and ‡ marks the further 11 that fall. The 5 cells in which an operation is wrong on all one hundred pairs carry neither mark: they survive in both units. Counting ties inside the denominator instead of outside leaves the same 35 cells and the same 15 operations. Removing the two types aligned with an operation's own attribute, darkening and desaturation, leaves all 15 operations still inverting; removing white noise instead leaves 9. Dropping one segment at a time, all 15 keep inverting in every one of the 20 rounds. **Ties are unevenly spread**: 4 operations tie on at least one pair and the largest count is 81. Where ties are many, accuracy below 50 per cent does not by itself mean the degraded copy is preferred; the reading that does is correct against incorrect, which is what the sign-flip test compares. Dk = darkening, Mo = motion blur, Bl = Gaussian blur, H2 = H.264, Rs = rescaling, St = desaturation, Ns = white noise, Jp = JPEG.

Operation	Types	Worst	Cells below chance (%)
Saliency area	6	**10.0**	Mo 29.0†, Bl 20.0†, Rs 21.0†, St 18.0‡, Ns 13.1‡, Jp 10.0‡
Hue count (Ke)	5	**4.2**	Dk 21.7†, Mo 15.7†, Bl 12.3‡, Rs 14.1†, St 4.2‡
Gradient orientation	4	**0.0**	Bl 26.0†, Rs 9.0, St 21.0†, Ns 0.0
Gradient magnitude	3	**0.0**	St 23.0†, Ns 0.0, Jp 5.0
Saturation	3	**1.0**	Dk 13.0‡, H2 15.0‡, Ns 1.0
Entropy	2	**0.0**	H2 10.0‡, Ns 0.0
Contrast (Ke)	2	**10.3**	St 31.6†, Ns 10.3‡
Edge distribution	2	**11.0**	H2 11.0‡, Jp 31.0†
Grey-level dispersion	2	**22.0**	St 22.0†, Ns 25.0†
GLCM contrast	1	**0.0**	Ns 0.0
Laplacian variance	1	**0.0**	Ns 0.0
Gygli mean of three	1	**3.0**	Ns 3.0
Colourfulness (Hasler)	1	**3.0**	Ns 3.0
Colourfulness (Ke)	1	**3.0**	Ns 3.0
Srinivas composite	1	29.0	Ns 29.0‡

Table S28 The SIFT+BoVW operation measured twice on the same 800 pairs: the reported run and an independent reimplementation of the same published operation under the same protocol. **The failure reproduces**: both collapse on white noise and JPEG, and both stay below the weakest published no-reference model at 80.8 per cent. They differ on desaturation, where the reported run inverts and the reimplementation does not, which is the one place where implementation detail changes a verdict. Red marks a cell below the 50 per cent chance floor.

Run	Darken	Motion	Blur	H.264	Rescale	Satur.	Noise	JPEG	Mean
Reported	100	100	90	90	88	**39**	**8**	**6**	**65.1**
Independent reimplementation	100	100	92	94	93	56	**10**	**6**	**68.9**

Table S29 *Source works* lists the archive citekeys whose full text matches the terms of each combination. **A dagger marks the works whose use was verified against the source with a recorded excerpt**, held in `o_que_cada_fonte_valida.json` of the reproducibility package; the others are retrieval matches and are not evidence that the work adopts that combination, so the rows without a dagger are evaluated here as diagnostic configurations rather than as published algorithms. Screening of 19 diagnostic configurations built from attributes identified in the archive; **the aggregations are defined in this study**, not taken from a published algorithm. *LBP* is the local binary pattern. *Sens.* is the within-reference mean against the degradation level on KADID-10k under the z-score aggregation and *Types* the count above .50 of 25. *Gen.* (z) is the direction-aligned correlation on KonIQ-10k as a magnitude, in the same configuration as *Sens.*; *Gen.* (ridge) is the generous variant, fitted outside the reference and reported separately. **Orientation comes from the data**, under the polarity-permissive rule of Section 3.3 of the main text, so a single-attribute row carries the opposite sign to Table S17 exactly when the declared direction contradicts the data. Eligibility uses the z-score configuration throughout; green marks the 1 row that clears the sensitivity gate.

Combination	Source works	Sens.	Types	Gen. (z)	Gen. (ridge)	Eligible
colourfulness + Laplacian variance + LBP uniformity	[12]†	**.524**	17	.061	.067	no
Laplacian variance + LBP uniformity	[74]	.439	15	.080	.082	no
edge distribution	[40], [84]	.298	9	.099	.099	no
spatial information	[43], [80], [81], [85], [100]	.251	13	.513	.513	no
LBP uniformity + gradient magnitude	[56]	.189	8	.177	.007	no
colourfulness	[23], [25]†	.182	9	.007	.007	no
LBP uniformity	[77], [8], [7], [22]	.175	12	.205	.205	no
colourfulness + exposure	[17]	.173	10	.243	.003	no
edge distribution + hue count + Laplacian variance	[42]†	.156	7	.204	.347	no
colourfulness + edge distribution + hue count + saliency	[78]†, [51]	.148	7	.162	.117	no
exposure + Laplacian variance	[44]†	.108	9	.500	.357	no
colourfulness + gradient magnitude	[70]†	.108	11	.296	.066	no
exposure	[29]†	.102	4	.343	.343	no
GLCM contrast + entropy	[90]†	.092	11	.371	.084	no
LBP uniformity + saliency	[69], [97], [95]	.089	9	.071	.277	no
saliency	[30], [57], [58], [68], [94], [93]	.075	3	.333	.333	no
Laplacian variance	[35], [53], [54]	.047	9	.601	.601	no
colourfulness + edge distribution	[89]†	.017	4	.085	.051	no
LBP uniformity + entropy	[2]	.009	8	.007	.285	no

Table S30 How the audited operations were identified, and what the protocol does and does not reach. The operations were not assembled from a database query, so the population of inference is the archive named in the first row and every count in this table is a count within it; no claim of exhaustive coverage is made. **The search instrument is itself measured against a manual reading, with the direction of the residual bias stated**, because a search whose accuracy is unknown cannot bound what it missed.

Step	What was done	What it yielded
Corpus	Curated full-text archive on video and multi-document summarization and on image-quality assessment, searched on 5 September 2026. **Not a database query, so the population of inference is this archive.**	186 PDFs
Patterns	One specific expression per operation family, never a bare word, so that *contrast* alone does not match. Reference lists are cut before matching; a cut leaving under 40 % of the PDF is refused and the work enters whole. Spatial information counts only when its context window carries ITU, the SI/TI pair, *index* or Sobel.	13 patterns
Instrument accuracy	Measured against a manual reading of the sources. **The reading and the inventory were done by the same author**, so the estimates may be optimistic. **The assessed articles are not a random sample of the archive**: they are the source works of the inventory together with articles selected because their combinations came closest to passing, and the second group inflates false positives by construction, so precision is not an unbiased estimate for the archive as a whole. Cutting reference lists raised precision from .47 without moving recall.	recall .81, precision .63, on 21 articles
Inventory inclusion rule	An operation enters only if its source declares a direction for its own statistic and scores a single frame. **No operation without a declared direction was screened in.** The 3 that declare a direction and stay out are sequence-level: `qaroush_movimento`, `qaroush_ses`, `sobel_cv`.	15 of 24 mapped operations
Coverage achieved	Share of the operation mentions in the archive that the screened set covers. A mention is not an article: one PDF may name several.	41 of 71 mentions (57.7 %)
What is left out, and why	**Of the 30 mentions outside, 25 were excluded under the inventory inclusion rule**, because the source specifies no direction (14) or no combination rule (11).	25 of 30 excluded
What the archive does instead	The term sweep is an automated retrieval check over an archive assembled during the thesis research; it is not a systematic mapping, and a match implies no reading of the matched work. Works citing no dedicated no-reference quality model, and how many of those match the patterns of one of the audited operations. **These are term matches, not confirmed use**: they were not read individually, and the same search precision applies here.	173 of 186 cite none; 33 of those matched an operation pattern

Table S31 Class-term matches in the curated summarization archive. It lists, work by work, which class terms the sweep found. A marker means the term sweep found the class in the body of the paper, with the bibliography removed. **This is triage, not a verdict**: a marker is a candidate for reading rather than a claim about the work. The term sweep is an automated retrieval check over an archive assembled during the thesis research; it is not a systematic mapping, and a match implies no reading of the matched work. The precision measured for the archive sweep recorded in Table S30 was measured on a different term set and is not claimed for this sweep. Class 4 is marked only when the text names the selection use, distance to a cluster centre or representativeness; cosine similarity alone is not counted, because in summarization it serves redundancy and clustering far more often than quality.

Work	C1	C2	C3	C4
[9]	•	•		•
[2]	•	•		•
[34]	•	•		•
[37]	•	•		•
[69]	•	•		•
[74]	•	•		•
[89]	•		•	•
[90]	•	•		•
[5]	•			•
[77]	•	•		
[8]	•	•		
[7]	•	•		
[11]	•	•		
[10]	•	•		
[12]	•	•		
[13]	•			•
[16]		•		•
[21]	•	•		
[22]	•	•		
[23]	•			•
[24]	•			•
[30]	•	•		
[43]	•	•		
[52]	•			•
[56]	•	•		
[60]	•	•		
[66]	•	•		

Work	C1	C2	C3	C4
[70]	•	•		
[71]	•	•		
[73]	•	•		
[76]		•		•
[78]	•			•
[82]	•	•		
[84]	•	•		
[87]	•			•
[85]	•			•
[88]	•	•		
[97]	•			•
[94]	•	•		
[6]				•
[20]	•			
[26]		•		
[28]	•			
[29]	•			
[33]		•		
[36]	•			
[38]		•		
[39]		•		
[44]	•			
[46]				•
[49]		•		
[57]	•			
[58]	•			
[59]				•
[64]	•			
[61]		•		
[68]	•			
[72]		•		
[75]		•		
[83]		•		
[96]		•		
[93]	•			
[95]	•			
[99]				•
[100]	•			
[101]	•			
[3]				
[4]				
[19]				
[41]				

Work	C1	C2	C3	C4
[45]				
[50]				
[48]				
[47]				
[1]				
[55]				
[65]				
[62]				
[63]				
[67]				
[14]				
[86]				
[91]				
[92]				
[98]				

Inferential procedure in full

Units of analysis, and which one each result uses.

Observations are nested and the unit is reported per analysis rather than once for the design. Strict accuracy on the constructed pairs is block-bootstrapped by topic; the paired score difference by segment; KADID sensitivity by reference image; the application by news topic. Observations that share content or a source enter the same block, which is the segment for pairs cut from one segment and the topic for clusters drawn from one news topic. All bootstraps use 10^4 resamples and seed 42. Table S26 recomputes every interval under the rival unit, and Table S27 reports per-cell significance under both the segment and the topic block.

Exact tests.

Per-cell significance uses an exact sign-flip randomization whose null enumerates all 2^b block sign assignments for b blocks, with Holm over the 120 operation–type tests. Disagreement counts between the complete-operation and matched-input regimes are compared by an exact McNemar test over the 42 paired clusters and, because those clusters come from 19 topics and McNemar treats them as independent, by the same block sign-flip over topics, which enumerates all 512 assignments the discordant pairs admit across 9 topics.

External association.

Spearman correlation against MOS or DMOS. Confidence intervals for KonIQ and CLIVE use the Fisher transform; KADID intervals and paired differences use reference-block bootstrap. Dependent correlations on the same images use Steiger's statistic [79]. The operational equivalence margin is $|\Delta\rho^*| < .02$, chosen as the largest association difference treated as practically negligible for this diagnostic rather than as a literature-wide standard; the paired difference and interval are reported so stricter margins can be applied. The multiplicity family contains all 630 pairwise contrasts among 36 evaluated scores (17 methods and 19 ablations); Bonferroni correction gives $\alpha = 0.05/630$ and critical $|z| = 3.95$. The accuracy, consistency and linearity triad after the monotonic mapping required by ITU-T P.1401 is in Table S4 [32].

The application estimand.

For each operation the application selects one representative per cluster. If y_c, l_c and h_c are the selected, lowest and highest reference scores in cluster c, and $\bar{y} = |C|^{-1} \sum_{c \in C} y_c$ with $\bar{l}$ and $\bar{h}$ defined analogously, the estimand is $100(\bar{y}-\bar{l})/(\bar{h}-\bar{l})$: a ratio of means, not an average of independently rescaled clusters. Zero-range clusters are undefined for per-cluster position and are counted explicitly. **The primary interval resamples the 18 source-aware blocks that the 19 topic labels of the application collapse to** and retains the clusters of each block: the two labels that share source videos count as one. This is the same grouping *rule* the constructed pairs use, on different data, so it yields a different *number* of blocks there (twelve, from thirteen labels). Two sensitivity analyses accompany it, one resampling the 19 separate topic labels and one resampling clusters within each sampled label. Raw branch-score differences, mean

per-cluster position, selected-maximum rate, range distributions and exclusions of the smallest 10% and 25% of ranges are in Table S10.

Degradation parameters.

Eight degradations at five levels: Gaussian blur, white noise, darkening, JPEG compression, low-bitrate H.264 transcoding, a downscale-and-restore round trip at five factors, horizontal motion blur and progressive desaturation, the last using $S'_\ell = \mathrm{clip}(\alpha_\ell S, 0, 255)$ with $\alpha_\ell \in \{0.75, 0.55, 0.35, 0.18, 0.05\}$. H.264 is applied to the frame sequence; the others are frame-based. Exact parameters are in Table S2.

Environment.

Experiments ran on one NVIDIA RTX 4060 GPU (8 GB). NR-IQA implementations use the PyIQA toolbox [15]; SIFT uses OpenCV and the vocabulary uses mini-batch K-means with $K = 300$ and `random_state=42`. Q-Align is quantized to 4-bit NormalFloat with double quantization and FP16 compute to fit the GPU [18]; no numerical equivalence to unquantized inference is assumed. Frames are decoded once and cached, and latency is re-measured in one software environment after ten warm-up and sixty timed iterations.

Table S32 The seventeen operations of the auxiliary comparison in Section 4.2 of the main text. ***Accuracy* is the same quantity the source-level synthesis (Section 4.7) of the main text reports**: strict accuracy in per cent, ties counted as errors, averaged over all eight transformation types, on the 800 original–transformed comparisons. *Aligned-excluded* is a second aggregation of the same measurement, reported here because it appears elsewhere in this supplement: it drops ties from the denominator and averages only over the types not aligned by construction with the operation, which differ from row to row, so it rests on the smaller number of comparisons in the last column. *Excluded* names those types and *Pairs* counts the comparisons they leave, **before ties are removed**, so the denominator that aggregation actually uses is smaller again wherever the operation ties. **The two aggregations are not interchangeable**: relative to the canonical column the second is higher for some operations, lower for others and equal where neither ties nor alignment applies. Both columns are derived from the same pair-level measurements. A double dagger marks the two sequence operations, which the isolated-frame inclusion rule excludes from the directional inventory and which enter only this auxiliary comparison. Under the canonical aggregation 8 of the seventeen fall below SIFT+BoVW at 65.1% and the median is 66.0%; over the 15 of the directional inventory alone the count is 6 and the median 74.8%. Identifiers in `typewriter` are function names in the released package, kept verbatim so each row can be matched to the code that produced it.

Identifier	Operation	Acc.	Aligned-ex.	Excluded	Pairs
`qaroush_saliencia`	Saliency	33.4	33.5	—	800
`ke_hue_count`	Hue count (Ke)	36.8	39.0	—	800
`qaroush_go`	Gradient orientation	53.9	53.9	—	800
`sobel_cv`‡	Sobel variation coefficient	59.6	56.3	Bl, Mo	600
`ke_edge_distribution`	Edge distribution	63.4	64.3	Bl, Mo	600
`qaroush_movimento`‡	Motion	63.5	61.7	Mo	700
`qaroush_entropia`	Entropy	63.8	63.8	—	800
`huang_saturacao`	Saturation	64.9	59.9	St	700
`qaroush_gm`	Gradient magnitude	66.0	66.0	—	800
`ke_contraste`	Contrast (Ke)	74.8	72.2	Dk	700
`laplacian_var`	Laplacian variance	76.4	68.5	Bl, Mo	600
`contraste_variancia`	Grey-level dispersion	79.6	76.7	Dk	700
`glcm_contraste`	GLCM contrast	79.9	77.0	Dk	700
`gygli_media_crua`	Averaged aesthetics (Gygli)	83.8	67.5	Bl, Dk, Mo, St	400
`hasler_colorfulness`	Colourfulness (Hasler)	85.2	83.1	St	700
`ke_colorfulness`	Colourfulness (Ke)	85.2	83.1	St	700
`srinivas_quality`	The mean rule (Srinivas)	87.0	74.5	Bl, Dk, Mo, St	400

Table S33 Replacing the score by a simpler quantity, on the 42 application clusters and 83 distinct segments. ρ is the Spearman correlation with the published score; *Same choice* counts the clusters whose selected representative is unchanged when the quantity replaces the score, and *Same order* the ordered pairs it preserves. **The descriptor count reproduces every choice**, so the operation is that count. **How many key frames the selector retains does not**: it tracks the score at +.177 and reproduces barely more than half the choices, while keypoints per frame tracks it at +.599. The count is therefore carried by detail per frame rather than by the number of frames. Segment duration ranges from 8.1 to 187.6 seconds and does not predict the score ($\rho = +.165$) or the count ($\rho = +.175$). Marks: * $p < .05$, ^{ns} not significant; all other rows have $p < .01$.

Quantity	What it is	ρ	Same choice	Same order
Descriptor count	the number of SIFT descriptors the key frames yield	+.995	42 / 42	63 / 63
Keypoints per frame	descriptor count divided by frames retained	+.599	29 / 42	44 / 63
Frames retained	how many key frames the selector keeps	$+.177^{\text{ns}}$	23 / 42	39 / 63
Score per frame	the score divided by frames retained	+.722	—	—
Score per descriptor	the mean rarity weight of Eq. 3 of the main article	$+.115^{\text{ns}}$	—	—

References

[1] (2024) Videogenic: Identifying highlight moments in videos with professional photographs as a prior. In: ACM Creativity and Cognition (C&C), https://doi.org/10.1145/3635636.3656186

[2] (2025) Contextual feature fusion-based keyframe selection using semantic attention and diversity-aware optimization. Symmetry 17(10):1737. https://doi.org/10.3390/sym17101737

[3] Apostolidis E, Adamantidou E, Metsai AI, et al (2021) Video summarization using deep neural networks: A survey. Proceedings of the IEEE 109(11):1838–1863. https://doi.org/10.1109/JPROC.2021.3117472

[4] Argaw DM, Yoon S, Caba Heilbron F, et al (2024) Scaling up video summarization pretraining with large language models. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 8332–8341, https://doi.org/10.1109/CVPR52733.2024.00796

[5] de Avila SEF, Lopes APBa, da Luz Jr. A, et al (2011) VSUMM: A mechanism designed to produce static video summaries and a novel evaluation method. Pattern Recognition Letters 32(1):56–68

[6] Barbara M, Maalouf A (2025) Prompts to summaries: Zero-shot language-guided video summarization. arXiv, URL https://arxiv.org/abs/2506.10807

[7] Barbieri TTdS (2021) Sumarização automática multivídeo baseada em estratégias humanas. PhD thesis, Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos, https://doi.org/10.11606/T.55.2021.tde-27092021-101106

[8] Barbieri TTdS, Goularte R (2021) Content selection criteria for news multi-video summarization based on human strategies. International Journal on Digital Libraries 22:1–14. https://doi.org/10.1007/s00799-020-00281-9

[9] Basavarajaiah M, Sharma P (2019) Survey of compressed domain video summarization techniques. ACM Computing Surveys

[10] Beserra AAR, Goularte R (2023) Multimodal early fusion operators for temporal video scene segmentation tasks. Multimedia Tools and Applications 82:31539–31556. https://doi.org/10.1007/s11042-023-14953-6

[11] Beserra AAR, Kishi RM, Goularte R (2020) Evaluating early fusion operators at mid-level feature space. In: Proc. Brazilian Symposium on Multimedia and the Web (WebMedia), pp 113–120, https://doi.org/10.1145/3428658.3431079

[12] Cakmak MC, Agarwal N, Poudel D (2025) TriPSS: A tri-modal keyframe extraction framework using perceptual, structural and semantic representations. ArXiv:2506.05395v2

[13] Chasanis V, Likas A, Galatsanos N (2008) Efficient video shot summarization using an enhanced spectral clustering approach. In: Artificial Neural Networks (ICANN)

[14] Chaves JMR, Tripathi S (2024) Videosage: Video summarization with graph representation learning. In: CVPR Workshop SG2RL, p s.p., URL https://arxiv.org/abs/2404.10539

[15] Chen C, Mo J (2022) IQA-PyTorch: PyTorch toolbox for image quality assessment. GitHub, URL https://github.com/chaofengc/IQA-PyTorch

[16] Chu WS, Song Y, Jaimes A (2015) Video co-summarization: Video summarization by visual co-occurrence. In: CVPR, pp 3584–3592, https://doi.org/10.1109/CVPR.2015.7298981, URL https://openaccess.thecvf.com/content_cvpr_2015/html/Chu_Video_Co-Summarization_Video_2015_CVPR_paper.html

[17] Datta R, Joshi D, Li J, et al (2006) Studying aesthetics in photographic images using a computational approach. In: Computer Vision – ECCV 2006, Lecture Notes in Computer Science, vol 3953. Springer, pp 288–301

[18] Dettmers T, Pagnoni A, Holtzman A, et al (2023) QLoRA: Efficient finetuning of quantized LLMs. In: NeurIPS, arXiv:2305.14314

[19] Duygulu P, Pan JY, Forsyth DA (2004) Towards auto-documentary: Tracking the evolution of news stories. In: Proceedings of the 12th annual ACM international conference on Multimedia, https://doi.org/10.1145/1027527.1027719

[20] Fu Y, Guo Y, Zhu Y, et al (2010) Multi-view video summarization. IEEE Transactions on Multimedia pp 717–729. https://doi.org/10.1109/TMM.2010.2052025, URL https://doi.org/10.1109/TMM.2010.2052025

[21] Gharbi H, Bahroun S, Massaoudi M, et al (2017) Key frames extraction using graph modularity clustering for efficient video summarization. In: IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)

[22] Gharbi H, Bahroun S, Zagrouba E (2019) Key frame extraction for video summarization using local description and repeatability graph clustering. Signal, Image and Video Processing 13(3):507–515

[23] Gygli M, Grabner H, Riemenschneider H, et al (2014) Creating summaries from user videos. In: Computer Vision – ECCV 2014, Lecture Notes in Computer Science, vol 8695. Springer, pp 505–520, https://doi.org/10.1007/978-3-319-10584-0_33

[24] Gygli M, Grabner H, Riemenschneider H, et al (2014) Creating summaries from user videos. In: ECCV, pp 505–520, https://doi.org/10.1007/978-3-319-10584-0_33

[25] Hasler D, Süsstrunk S (2003) Measuring colourfulness in natural images. In: Human Vision and Electronic Imaging VIII, pp 87–95, https://doi.org/10.1117/12.477378

[26] Hong R, Tang J, Tan HK, et al (2011) Beyond search: Event-driven summarization for web videos. ACM Transactions on Multimedia Computing, Communications, and Applications 7(4):1–18. https://doi.org/10.1145/2043612.2043613

[27] Hou X, Zhang L (2007) Saliency detection: A spectral residual approach. In: 2007 IEEE Conference on Computer Vision and Pattern Recognition, pp 1–8, https://doi.org/10.1109/CVPR.2007.383267

[28] Hua H, Tang Y, Xu C, et al (2025) V2xum-llm: Cross-modal video summarization with temporal prompt instruction tuning. In: AAAI, p s.p., URL https://arxiv.org/abs/2404.12353

[29] Huang H, Wu Z, Pang G, et al (2024) An aesthetic-driven approach to unsupervised video summarization. IEEE Access 12. https://doi.org/10.1109/ACCESS.2024.3434508

[30] Hussain T, Muhammad K, Ding W, et al (2021) A comprehensive survey of multi-view video summarization. Pattern Recognition p 107567. https://doi.org/10.1016/j.patcog.2020.107567, URL https://doi.org/10.1016/j.patcog.2020.107567

[31] International Telecommunication Union (2008) Subjective video quality assessment methods for multimedia applications. ITU-T Recommendation P.910

[32] International Telecommunication Union (2020) Methods, metrics and procedures for statistical evaluation of objective quality prediction models. ITU-T Recommendation P.1401

[33] Islam MM, Kakouros S, Heikkilä J, et al (2025) Towards an automated multimodal approach for video summarization: Building a bridge between text, audio and facial cue-based summarization. In: Proceedings of the Workshop

[34] Jeong Dj, Yoo HJ, Cho NI (2017) A static video summarization method based on the sparse coding of features and representativeness of frames. EURASIP Journal on Image and Video Processing 2017(1):1. https://doi.org/10.1186/s13640-016-0122-9

[35] Jeong DJ, Yoo HJ, Cho NI (2017) A static video summarization method based on the sparse coding of features and representativeness of frames. EURASIP Journal on Image and Video Processing 2017:1. https://doi.org/10.1186/s13640-016-0122-9, URL https://doi.org/10.1186/s13640-016-0122-9

[36] Ji Z, Ma Y, Pang Y, et al (2018) Hypergraph dominant set based multi-video summarization. Signal Processing pp 114–123. https://doi.org/10.1016/j.sigpro.2018.01.028, URL https://doi.org/10.1016/j.sigpro.2018.01.028

[37] Ji Z, Ma Y, Pang Y, et al (2019) Query-aware sparse coding for web multi-video summarization. Information Sciences pp 274–289. https://doi.org/10.1016/j.ins.2018.09.050, URL https://arxiv.org/abs/1707.04021

[38] Kadam P, Vora D (2025) Systematic frame selection and quality assessment for efficient video summarization. Multimedia Systems 31(4):279. https://doi.org/10.1007/s00530-025-01860-z

[39] Kalkhorani V, et al (2024) Beyond the frame: Single and multiple video summarization. arXiv, URL https://arxiv.org/abs/2401.10254

[40] Kannao R, Guha P (2020) A system for semantic segmentation of TV news broadcast videos. Multimedia Tools and Applications 79(9–10):6191–6225. https://doi.org/10.1007/s11042-019-08445-9

[41] Kaushal V, Kothawade S, Iyer R, et al (2020) Realistic video summarization through VISIOCITY: A new benchmark and evaluation framework. In: Proceedings of the 2nd International Workshop on AI for Smart TV Content Production, Access and Delivery (AI4TV), ACM MM, https://doi.org/10.1145/3422839.3423064

[42] Ke Y, Tang X, Jing F (2006) The design of high-level features for photo quality assessment. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp 419–426, https://doi.org/10.1109/CVPR.2006.303

[43] Kishi RM, Trojahn TH, Goularte R (2019) Correlation based feature fusion for the temporal video scene segmentation task. Multimedia Tools and Applications 78(11):15623–15646. https://doi.org/10.1007/s11042-018-6959-4

[44] Korolkov V (2025) Scene detection policies and keyframe extraction strategies for large-scale video analysis. ArXiv:2506.00667

[45] Lee MJ, Gong D, Cho M (2025) Video summarization with large language models. In: Proc. CVPR, pp 18981–18991, https://doi.org/10.1109/CVPR52734.2025.01768

[46] Lei Z, Zhang C, Zhang Q, et al (2019) FrameRank: A text processing approach to video summarization. In: IEEE International Conference on Multimedia and Expo (ICME)

[47] Li W, Han W, Deng LJ, et al (2025) Spiking variational graph representation inference for video summarization. IEEE Transactions on Image Processing pp 5697–5709. https://doi.org/10.1109/TIP.2025.3602649, URL https://arxiv.org/abs/2508.15389

[48] Li W, Han W, Man H, et al (2025) Language-guided graph representation learning for video summarization. arXiv / TPAMI accepted, URL https://arxiv.org/abs/2511.10953

[49] Li Y, Merialdo B (2010) Multi-video summarization based on Video-MMR. In: Proc. WIAMIS

[50] Li Y, Merialdo B (2012) Video summarization based on balanced av-mmr. In: MultiMedia Modeling, pp 370–382, https://doi.org/10.1007/978-3-642-27355-1_35, URL https://www.eurecom.fr/publication/3510

[51] Luo W, Wang X, Tang X (2013) Content-based photo quality assessment. IEEE Transactions on Multimedia 15(8):1930–1943. https://doi.org/10.1109/TMM.2013.2269899

[52] Messaoud S, Lourentzou I, Boughoula A, et al (2021) Deepqamvs: Query-aware hierarchical pointer networks for multi-video summarization. In: SIGIR, pp

1389–1399, https://doi.org/10.1145/3404835.3462959, URL https://arxiv.org/abs/2105.06441

[53] Mittal A, Moorthy AK, Bovik AC (2012) No-reference image quality assessment in the spatial domain. IEEE Transactions on Image Processing 21(12):4695–4708. https://doi.org/10.1109/TIP.2012.2214050

[54] Mittal A, Soundararajan R, Bovik AC (2013) Making a "completely blind" image quality analyzer. IEEE Signal Processing Letters 20(3):209–212. https://doi.org/10.1109/LSP.2012.2227726

[55] Moraes L, Marcacini RM, Goularte R (2022) Video summarization using text subjectivity classification. In: Proc. WebMedia, pp 141–149, https://doi.org/10.1145/3539637.3556998

[56] Nandini HM, Chethan HK, Rashmi BS (2022) Shot based keyframe extraction using edge-lbp approach. Journal of King Saud University - Computer and Information Sciences 34(7):4537–4545. https://doi.org/10.1016/j.jksuci.2020.10.031

[57] Ngo CW, Ma YF, Zhang HJ (2005) Video summarization and scene detection by graph modeling. IEEE Transactions on Circuits and Systems for Video Technology 15(2):296–305

[58] Nie L, Hong R, Zhang L, et al (2016) Perceptual attributes optimization for multivideo summarization. IEEE Transactions on Cybernetics 46(12):2991–3003. https://doi.org/10.1109/TCYB.2015.2493558, URL https://doi.org/10.1109/TCYB.2015.2493558

[59] Otani M, Nakashima Y, Rahtu E, et al (2019) Rethinking the evaluation of video summaries. In: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 7588–7596, https://doi.org/10.1109/CVPR.2019.00778

[60] Over P, Smeaton AF, Awad G (2008) The TRECVid 2008 BBC rushes summarization evaluation. In: Proceedings of the 2nd ACM TRECVid Video Summarization Workshop (TVS '08). ACM, https://doi.org/10.1145/1463563.1463564

[61] Panda R, Roy-Chowdhury AK (2017) Collaborative summarization of topic-related videos. In: Proc. CVPR, pp 4274–4283, https://doi.org/10.1109/CVPR.2017.455

[62] Panda R, Roy-Chowdhury AK (2017) Multi-view surveillance video summarization via joint embedding and sparse optimization. IEEE Transactions on Multimedia pp 2010–2021. https://doi.org/10.1109/TMM.2017.2708981, URL https://arxiv.org/abs/1706.03121

[63] Panda R, Roy-Chowdhury AK (2017) Sparse modeling for topic-oriented video summarization. In: ICASSP, pp 1388–1392, https://doi.org/10.1109/ICASSP.2017.7952384, URL https://dblp.org/rec/conf/icassp/PandaR17

[64] Panda R, Das A, Roy-Chowdhury AK (2016) Embedded sparse coding for summarizing multi-view videos. In: IEEE ICIP, pp 191–195, https://doi.org/10.1109/ICIP.2016.7532345, URL https://rpand002.github.io/data/ICIP_2016.pdf

[65] Panda R, Das A, Roy-Chowdhury AK (2016) Video summarization in a multi-view camera network. ICPR / arXiv, URL https://arxiv.org/abs/1608.00310

[66] Panda R, Mithun NC, Roy-Chowdhury AK (2017) Diversity-aware multi-video summarization. IEEE Transactions on Image Processing 26(10):4712–4724. https://doi.org/10.1109/TIP.2017.2708902

[67] Park J, Lee J, Kim IJ, et al (2020) Sumgraph: Video summarization via recursive graph modeling. In: Lecture Notes in Computer Science, pp 647–663, https://doi.org/10.1007/978-3-030-58595-2_39, URL https://arxiv.org/abs/2007.08809

[68] Park J, Lee J, Sohn K (2025) Language-guided recursive spatiotemporal graph modeling for video summarization. International Journal of Computer Vision pp 8617–8641. https://doi.org/10.1007/s11263-025-02577-2, URL https://doi.org/10.1007/s11263-025-02577-2

[69] Qaroush AM, Jubran M, Olayan Q (2025) Graph-based centrality framework for effective multi-video summarization. Information Processing & Management p 104276. https://doi.org/10.1016/j.ipm.2025.104276, URL https://doi.org/10.1016/j.ipm.2025.104276

[70] Qaroush AM, Jubran M, Olayyan Q, et al (2025) Maximum relevant diversity aware, multi-video summarization using clustering and evolutionally multi-objective optimization. Expert Systems with Applications 292:128631. https://doi.org/10.1016/j.eswa.2025.128631

[71] Saqib S, Kazmi SAR (2018) Video summarization for sign languages using the median of entropy of mean frames method. Entropy 20(10):748. https://doi.org/10.3390/e20100748, URL https://doi.org/10.3390/e20100748

[72] Shao J, Jiang D, Wang M, et al (2010) Multi-video summarization using complex graph clustering and mining. Computer Science and Information Systems pp 85–98. https://doi.org/10.2298/CSIS1001085S, URL https://doiserbia.nb.rs/Article.aspx?id=1820-02141001085S

[73] Sidiropoulos P, Mezaris V, Kompatsiaris I, et al (2011) Temporal video segmentation to scenes using high-level audiovisual features. IEEE TCSVT 21(8):1163–1177. https://doi.org/10.1109/TCSVT.2011.2138830

[74] Singh M, Ooi WT (2026) Graph-based event and sub-event grouping in user-generated videos with distortion-aware keyframe clustering. ACM Transactions on Multimedia Computing, Communications and Applications

[75] Soares ER, Barrère E (2018) Automatic topic segmentation for video lectures using low and high-level audio features. In: Proceedings of the 24th Brazilian Symposium on Multimedia and the Web (WebMedia), pp 189–196, https://doi.org/10.1145/3243082.3243096

[76] Song Y, Vallmitjana J, Stent A, et al (2015) TVSum: Summarizing web videos using titles. In: Proc. CVPR, pp 5179–5187, https://doi.org/10.1109/CVPR.2015.7299154

[77] de Souza Barbieri TT, Goularte R (2020) Investigating subjectivity criterion for multi-video summarization. In: Proceedings of the Brazilian Symposium on Multimedia and the Web (WebMedia). ACM, pp 137–144, https://doi.org/10.1145/3428658.3430964

[78] Srinivas M, Pai MMM, Pai RM (2016) An improved algorithm for video summarization — a rank based approach. Procedia Computer Science 89:812–819. https://doi.org/10.1016/j.procs.2016.06.065

[79] Steiger JH (1980) Tests for comparing elements of a correlation matrix. Psychological Bulletin 87(2):245–251. https://doi.org/10.1037/0033-2909.87.2.245

[80] Swain MJ, Ballard DH (1991) Color indexing. International Journal of Computer Vision 7(1):11–32. https://doi.org/10.1007/BF00130487

[81] Taylor D, Caceres RS, Mucha PJ (2017) Super-resolution community detection for layer-aggregated multilayer networks. Physical Review X 7(3):031056

[82] Trojahn TH, Goularte R (2021) Temporal video scene segmentation using deep-learning. Multimedia Tools and Applications 80(12):17487–17513. https://doi.org/10.1007/s11042-020-10450-2

[83] Wang F, Merialdo B (2009) Multi-document video summarization. In: IEEE ICME, pp 1326–1329, https://doi.org/10.1109/ICME.2009.5202747, URL https://www.eurecom.fr/publication/2751

[84] Wang M, Hong R, Li G, et al (2012) Event-driven web video summarization by tag localization and key-shot identification. IEEE Transactions on Multimedia pp 975–985. https://doi.org/10.1109/TMM.2012.2185041, URL https://doi.org/10.1109/TMM.2012.2185041

[85] Wu G, Wang M, Ma N (2026) Multimodal video summarization based on graph contrastive learning with fine-grained graph interaction. Signal Processing p 110250. https://doi.org/10.1016/j.sigpro.2025.110250, URL https://doi.org/10.

1016/j.sigpro.2025.110250

[86] Wu J, hua Zhong S, Liu Y (2019) Mvsgcn: A novel graph convolutional network for multi-video summarization. In: ACM Multimedia, pp 827–835, https://doi.org/10.1145/3343031.3350938, URL https://research.polyu.edu.hk/en/publications/a-novel-graph-convolutional-network-for-multi-video-summarization/

[87] Wu J, hua Zhong S, Liu Y (2020) Dynamic graph convolutional network for multi-video summarization. Pattern Recognition p 107382. https://doi.org/10.1016/j.patcog.2020.107382, URL https://doi.org/10.1016/j.patcog.2020.107382

[88] Xie Xn, Wu F (2008) Automatic video summarization by affinity propagation clustering and semantic content mining. In: International Symposium on Electronic Commerce and Security

[89] Xu Y, Zheng J, Tao Y (2024) Property constrained video summarization via regret minimization. SN Computer Science 5:254. https://doi.org/10.1007/s42979-023-02588-1

[90] Yang G, He Z, Su Z, et al (2024) Keyframe recommendation based on feature intercross and fusion. Complex & Intelligent Systems 10:4955–4971. https://doi.org/10.1007/s40747-024-01417-z

[91] Yeh CH, Lien CM, Zhan ZX, et al (2025) Graph convolutional network for fast video summarization in compressed domain. Neurocomputing p 128945. https://doi.org/10.1016/j.neucom.2024.128945, URL https://doi.org/10.1016/j.neucom.2024.128945

[92] Yu Q, Wang Z, Wei G, et al (2026) Deep learning for video summarization: Systematic review, challenges and opportunities. IEEE/CAA Journal of Automatica Sinica 13(1):21–42. https://doi.org/10.1109/JAS.2025.125864

[93] Zhang L, Gao Y, Hong R, et al (2015) Probabilistic skimlets fusion for summarizing multiple consumer landmark videos. IEEE Transactions on Multimedia 17(1):40–49. https://doi.org/10.1109/TMM.2014.2370257

[94] Zhang L, Xia Y, Mao K, et al (2015) An effective video summarization framework toward handheld devices. IEEE Transactions on Industrial Electronics 62(2):1309–1316. https://doi.org/10.1109/TIE.2014.2336639

[95] Zhang Y, Zimmermann R (2016) Efficient summarization from multiple georeferenced user-generated videos. IEEE Transactions on Multimedia 18(3):418–431. https://doi.org/10.1109/TMM.2016.2520827

[96] Zhang Y, Ma H, Zimmermann R (2013) Dynamic multi-video summarization of sensor-rich videos in geo-space. In: Advances in Multimedia Modeling (MMM),

Lecture Notes in Computer Science, vol 7732. Springer, pp 380–390, https://doi.org/10.1007/978-3-642-35725-1_35

[97] Zhang Y, Zhang L, Zimmermann R (2015) Aesthetics-guided summarization from multiple user generated videos. ACM Transactions on Multimedia Computing, Communications, and Applications 11(2):24:1–24:23. https://doi.org/10.1145/2659520, URL https://doi.org/10.1145/2659520

[98] Zhao Z, Bai C, Hao P (2026) Graphvsum: Graph guided multimodal video summarization. Multimedia Systems p s.p. https://doi.org/10.1007/s00530-025-02084-x, URL https://doi.org/10.1007/s00530-025-02084-x

[99] Zhou K, Qiao Y, Xiang T (2018) Deep reinforcement learning for unsupervised video summarization with diversity-representativeness reward. In: Proc. AAAI Conference on Artificial Intelligence, pp 7582–7589

[100] Zhu X, Liu J, Wang J, et al (2014) Key observation selection-based effective video synopsis for camera network. Machine Vision and Applications 25:145–157. https://doi.org/10.1007/s00138-013-0519-8

[101] Zhu Y, Zhao W, Hua R, et al (2023) Topic-aware video summarization using multimodal transformer. Pattern Recognition 140:109578. https://doi.org/10.1016/j.patcog.2023.109578

Model eligibility and output quality

This section carries the model-level detail summarized in Sections 4.5 and 4.6 of the main text: how robust the eligible set is, the in-domain floor model by model, a common-reference comparison, and whether substituting the scorer improves the output according to DOVER.

Three methods are eligible: LIQE, QualiCLIP and CLIP-IQA+, and no single one is best on both axes. The set is unchanged when the generalization threshold varies from .60 to .65 or the coverage cut from 12 to 22 types; at .55 MUSIQ is added and at .70 only LIQE remains. The universal unseen-database clause is what separates them: TOPIQ, MUSIQ, DBCNN and HyperIQA are strong on the unseen authentic database and fall to between .462 and .556 on KADID. Part of the gap between SIFT+BoVW and the eligible methods on KADID is a change of regime: the SIFT+BoVW form adds a term proportional to $\log_{10}(K/N)$ in the descriptor count, positive over the small broadcast clusters and negative over a database of thousands of images, so moving to KADID changes how the score is built; the derivation of that term is given in this supplement.

On the in-domain floor the three pass with overlapping topic-block intervals, LIQE agreeing with Q-Align, the in-domain reference of the model set, on 32 of 42 clusters ($\kappa_{\text{uniform}} = .551$, 95% CI [.273, .792] over the 18 source-aware blocks). The SIFT+BoVW operation agrees on 21, only 1.3 decisions above the uniform expectation, and its interval crosses chance (.056, [−.302, .420]). Against LIQE, the reference the protocol assigns to the literature-derived operations, the same operation agrees on 20 of 42 ($\kappa_{\text{uniform}} = .011$), and that interval crosses chance as well. Several literature-derived operations meet this condition on unmodified broadcasts despite failing both external ones, and three lose that pass when one topic is removed. Keeping the 19 topic labels apart leaves both verdicts unchanged ([.284, .786] and [−.308, .412]).

An auxiliary comparison places every scorer against one common reference, and its group is not the complete-case candidate set. Because LIQE is the designated reference for the literature-derived operations, it cannot also be a candidate there, so the group comprises the nine other complete-case candidates and ARNIQA in its place. All ten meet the condition, with median normalized agreement .708 against .281 for the literature-derived operations.

The sensitivity axis alone already differs between the two benchmarks. Applied to the constructed broadcast pairs, the same rule keeps the mean threshold of .50 and rescales the coverage clause in proportion, from 15 of KADID's 25 distortion types to 5 of the 8 types built here; it admits six of the fifteen, and none meets it on both benchmarks. Cost is secondary but not negligible: the SIFT+BoVW operation takes 182 s per cluster and LIQE 1.9 s, a system-level figure that includes sampling and pooling.

Output quality with DOVER.

Its technical branch, DOVER-T, is the output measure. The selected segments are rewrapped as common 5-frame-per-second MP4 inputs and evaluated using the official view decomposition and paired seed; Table S12 specifies fragment sampling. For selected reference score y_c and the lowest and highest reference scores l_c, h_c in each

cluster, the achievable-range position is

$$100\,\frac{\bar{y}-\bar{l}}{\bar{h}-\bar{l}}, \qquad \bar{y}=|C|^{-1}\sum_{c\in C} y_c, \tag{1}$$

with analogous definitions for $\bar{l}$ and $\bar{h}$. This ratio of means is distinct from averaging separately normalized cluster positions. It is defined when $\bar{h}>\bar{l}$ and expresses range points on a reference model's scale, not a percentage of human quality.

On the complete substitution, the global achievable-range position of Equation (1), not the branch score itself, rises from 50.5% to 76.0%, +25.5 range points, but its block interval $[-13.5,+54.6]$ and hierarchical interval $[-17.2,+59.6]$ both cross zero, so no population-level effect is confirmed. Under matched inputs, with at most ten uniformly spaced frames and mean pooling for both, the DOVER-T contrast is +27.1 range points with both intervals positive (block $[5.2,46.6]$; hierarchical $[2.4,52.5]$), and +.0074 on the raw branch score. Block intervals merge the two topic labels that share source videos; hierarchical intervals resample clusters within the 19 separate labels, and no verdict differs under the separate-label grouping.